

High-End Timer Transfer Unit (HTU) Module

This chapter describes the high-end timer transfer unit (HTU) module. The HTU is similar to the DMA (Direct Memory Access) module, but it is specialized to transfer N2HET (High-End Timer) data to or from the microcontroller RAM.

NOTE: This chapter describes a superset implementation of the HTU module that includes features and functionality that require DMA. Since not all devices have DMA capability, consult your device-specific datasheet to determine applicability of these features and functions to your device being used.

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24.1 Overview

The HET transfer unit is a dedicated direct memory access controller that transfers data between the N2HET RAM and RAM buffers located in the main memory address range. This eliminates time consuming CPU accesses to the N2HET RAM to gather measurement data or creating output waveforms and thus freeing up the CPU to perform other tasks.

24.1.1 Features

- Independently transfers data between the N2HET and the main memory
- 8 double control packets supporting dual buffer configuration
- Transfer requests generated by N2HET instructions/events
- One shot, circular and auto switch buffer transfer modes for each double control packet for flexible buffer handling
- Constant and post-increment addressing modes
- 32- or 64-bit transactions
- Programmable memory protection region
- Parity protect control packet RAM
- Extensive diagnostic functionality

24.2 Module Operation

The HTU is tightly coupled to the N2HET and is not intended to transfer data from other peripheral modules. It initiates transfers with the help of requests generated by the N2HET program and configurable control packets. Figure 24-1 shows a system block diagram of the HTU and the main path for the data transfer. The tight coupling and the dedicated bus into the SCR (Switched Central Resource) reduces the amount of data transferred on the peripheral bus, which increases the overall system performance. However if the application decides to use the direct CPU access method to the N2HET RAM, it is free to do so.

Figure 24-2 shows a more detailed block diagram of the HTU module.

Figure 24-1. System Block Diagram

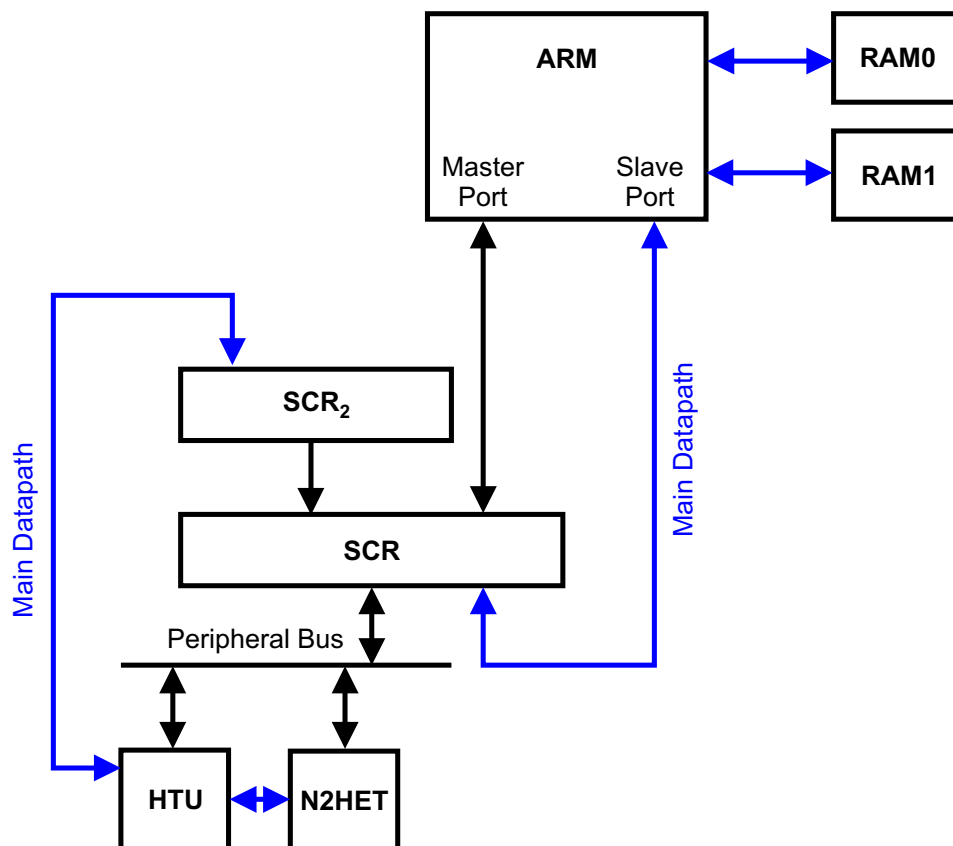
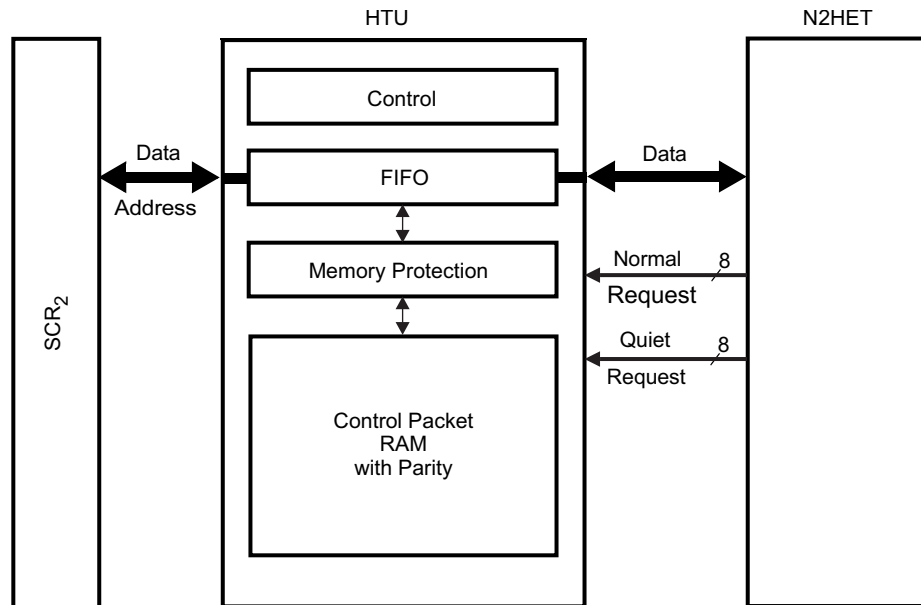


Figure 24-2. HTU Block Diagram


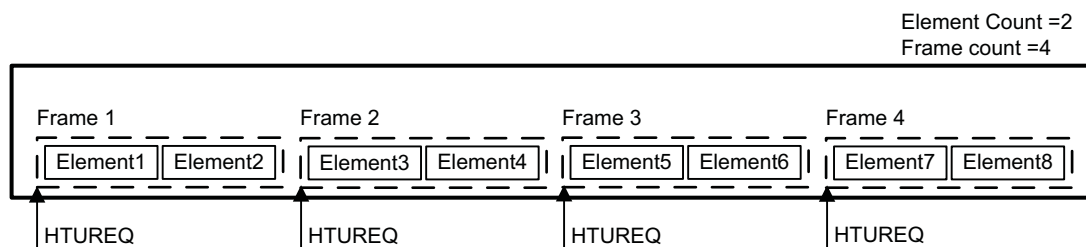
Transfers between N2HET RAM and the main memory are triggered by 8 different normal N2HET requests. Quiet requests are used for specific cases and are discussed in [Section 24.2.4.1](#). Control packets, which store the source and destination addresses, the transfer count and other information (see [Section 24.5](#)), are associated with the requests. A FIFO decouples the read- and write-path and allows to do data-packing in the case of different read- and write-data sizes. The application can specify a section of memory into or from which the data is transferred. This serves as memory protection in the case that information in the control packet RAM was unintentionally altered and avoids that the HTU can overwrite important application data.

Control packets are implemented as double control packets (DCP) which allow to specify two buffers for the data transfer. This enables the CPU to work with one buffer, while new data is transferred to/from the other buffer.

The control packet defines:

- the start address of the source/destination buffers
- the N2HET instruction address location
- how many elements need to be transferred per request
- the buffer size as the number of elements times the number of frames
- the buffer handling

A transfer is triggered when a certain condition (for example, capture, compare condition) is detected by a N2HET instruction. The N2HET instruction specifies which request line to the HTU will be triggered at the event. The DCPs have a fixed assignment to the request lines and the corresponding assignment can be found in the device datasheet. Once a request is triggered, it starts a frame transfer. A frame can contain one or more elements. Elements are defined as 32-bit or 64-bit words of data.

Figure 24-3. Example of a HTU Transfer


24.2.1 Data Transfers between Main RAM and N2HET RAM

24.2.1.1 Addressing Modes

The addressing modes of a control packet need to be distinguished between the main RAM of the CPU and the N2HET RAM.

Main RAM

For each double control packet (see [Section 24.2.1.3](#)), the addressing mode for the main RAM (RAM0/1) can be configured to constant or post-increment mode in register IHADDRCT.

- **Constant Addressing:** In constant mode, the HTU writes/reads the data to/from the same address in the main RAM.
- **Post-increment Addressing:** In post-increment mode, the HTU writes/reads the data to/from the main RAM by incrementing through the addresses after each transfer. If 32-bit transfers are selected it will automatically increment by 4 Byte, if 64-bit transfers are selected, it will increment by 8 Byte. The examples of Use Cases illustrate the post-increment mode, where the elements of consecutive frames are transferred to/from consecutive locations in the main RAM buffer.

N2HET RAM

How a DCP addresses the N2HET RAM is determined by the initial N2HET address, the initial element counter (IETCOUNT) and the N2HET addressing mode (ADDMMH). The main difference to the main RAM addressing mode is that the HET address is reset to the initial HET address for every first element of a frame. To implement constant addressing, the initial element counter needs to be set to 1. Post-increment addressing is selected by programming the initial element counter to a value other than 1.

24.2.1.2 Single Buffer Implementation

In a single buffer implementation, the DCP is set up to transfer data to/from a single buffer in the main RAM. With each transfer request, the programmed number of elements is transferred and the buffer pointer is reset to its starting address after the programmed number of frame transfers have completed.

[Figure 24-4](#) shows the request on one request line of the HTU and the frame running on the assigned control packet visualized by the element counter. In the diagram, the frame has 5 element transfers (element count = 5).

Before the application reads the buffer, it has to disable the control packet to avoid that new data overwrites the buffer while it's being accessed by the application. Regardless of the control packet being disabled at t1 or t2 the last frame will always be completed, since the trigger request has been received already. The application can determine any ongoing transfers by the TIPF flag and the NACP bits.

- **One Shot Buffer Mode:** If TMBA or TMBB is set to one shot buffer mode then the data stream will stop after all elements of buffer A or buffer B have been transferred. This means that the corresponding DCP will be disabled after the last frame was transferred to/from buffer A or B and CFTCTA or CFTCTB decrements to 0.
- **Circular Buffer Mode:** If TMBA or TMBB is set to circular buffer mode, then the data stream will continue back at the start of buffer A or B after all elements of buffer A or B have been transferred. The example of Timing Example for Circular Buffer Mode assumes IETCOUNT = 3 (Initial Element Transfer Count), IFTCOUNT = 3 (Initial Frame Transfer Count, SIZE = 0 (Size of Transfer = 32-bit) and ADDFM = 0 (Addressing Mode Main Memory = Post Increment). So there are in total 9 32-bit values in the buffer. It also assumes IFADDRx = 10h. "U" means uninitialized.

Figure 24-4. Single Buffer Timing and Memory Representation

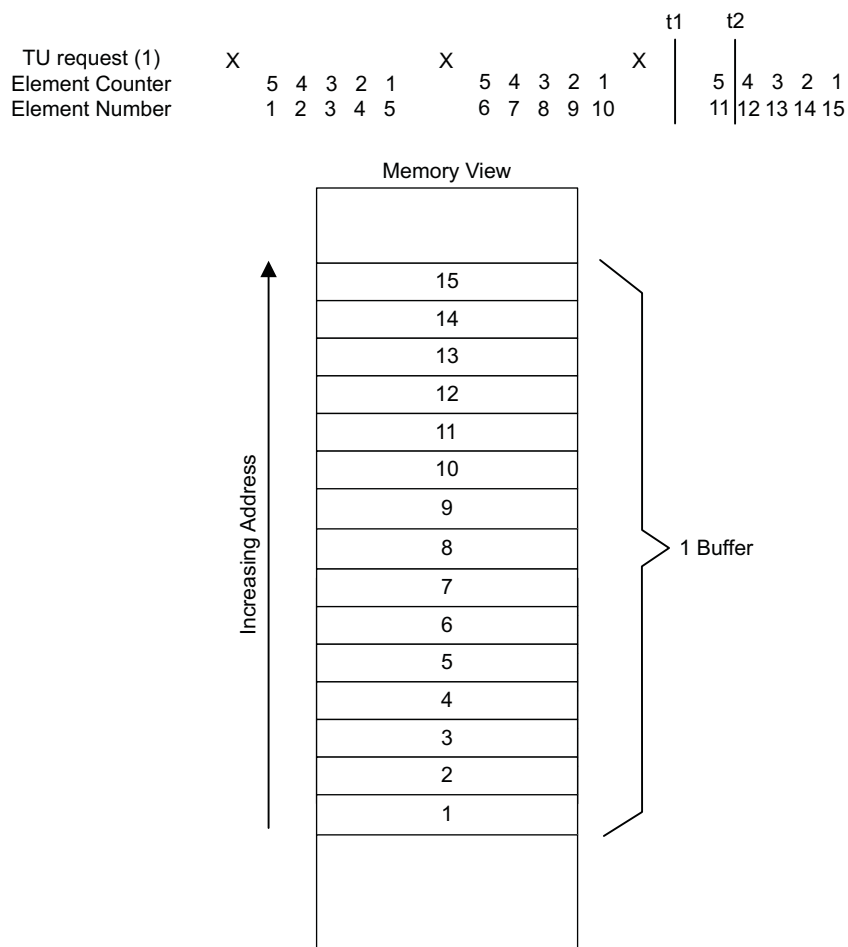
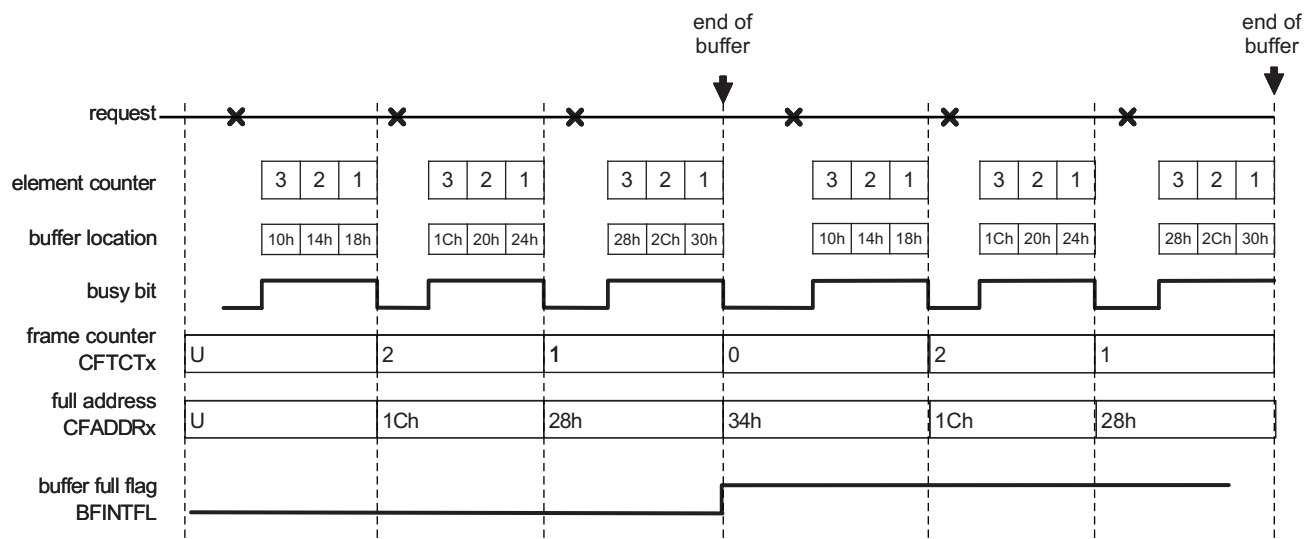


Figure 24-5. Timing Example for Circular Buffer Mode

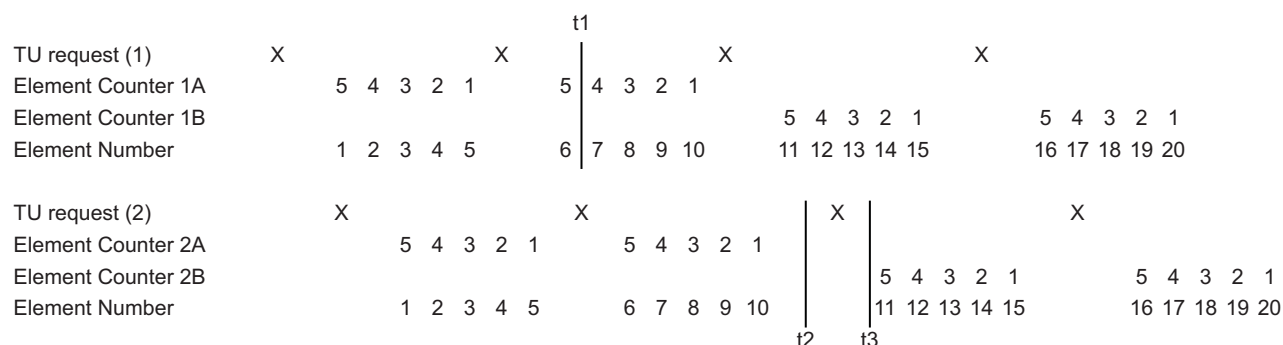


24.2.1.3 Dual Buffer Implementation

The transfer unit provides **double control packets (DCPs)** supporting the use of two buffers per data stream (per HTU request source). If one buffer should be read by the CPU or DMA, the data stream is directed to the other buffer and the first buffer is frozen. Switching to the other buffer can be triggered with a write access to the CPENA register or with the DCP configured to automatically switch to the other buffer when the programmed number of frames has been transmitted. Freezing the buffer avoids this buffer to be overwritten with new HET data while the CPU or DMA reads this buffer.

Figure 24-6 shows a timing example of two HET instructions 1 and 2, which are the request sources for the HTU (and are controlled by DCP 1 and DCP 2). Each generated frame has 5 element transfers. Request source 1 has two RAM buffers, controlled by two control packets 1A and 1B. Request source 2 has two RAM buffers, controlled by two control packets 2A and 2B.

Figure 24-6. Dual Buffer Timing



Memory View for DCP-1A/B

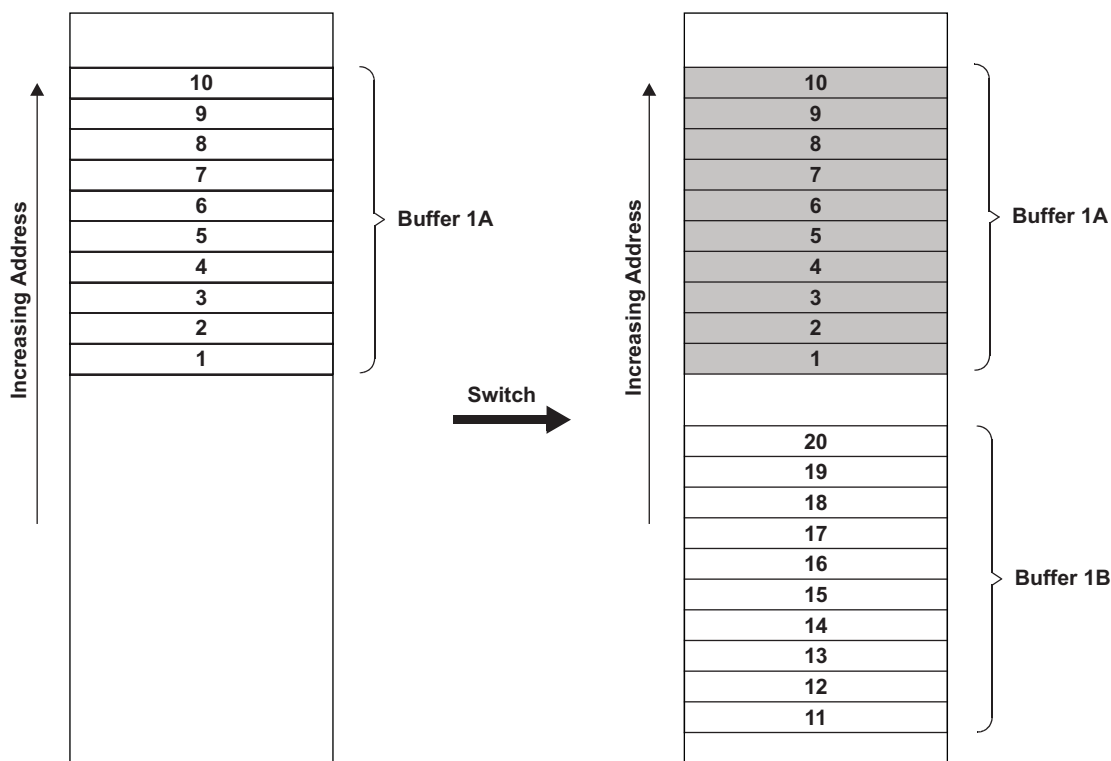


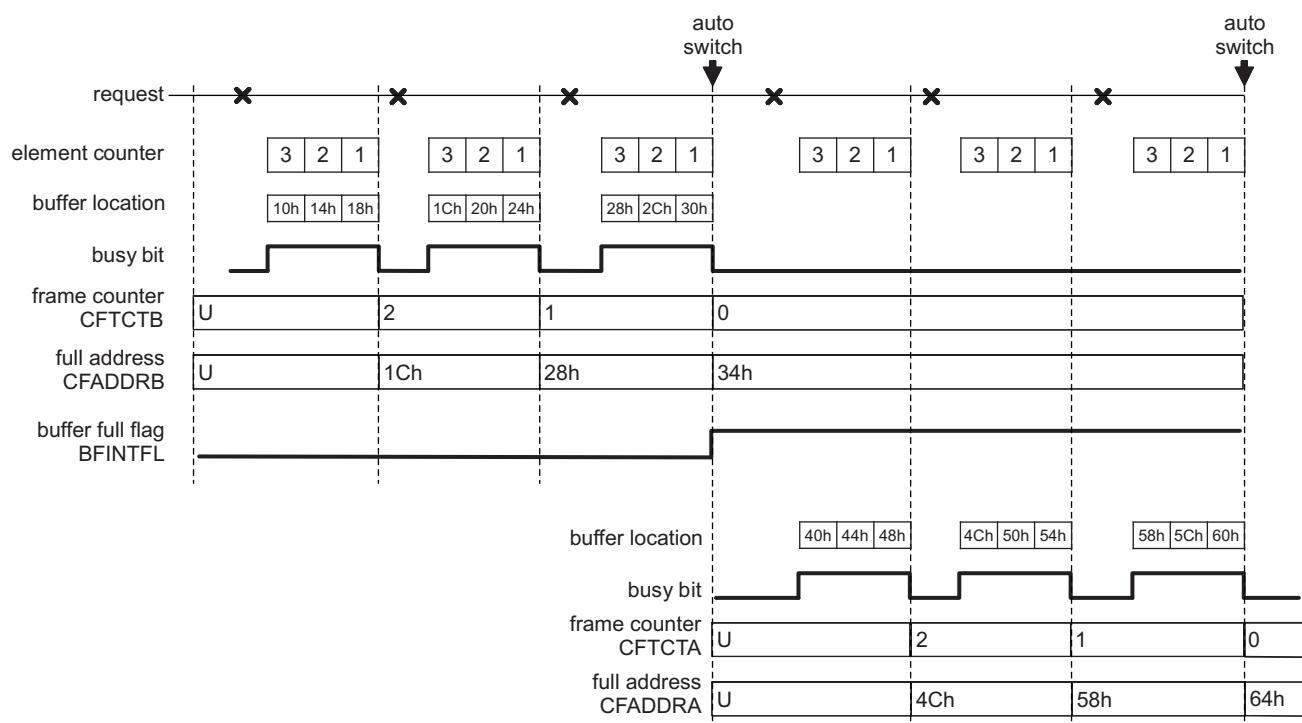
Figure 24-6 shows a switch at time t1, where buffer 1A is frozen and data stream 1 is directed to buffer 1B, but only after the frame has been completed. It also shows the time (t2 or t3) where 2A is frozen and data stream 2 is directed to buffer 2B. If the switch happens between the request and the start of the frame (for example, time t3), then the frame is processed by the new control packet (although the old control packet was active at the time of the request). The delays between the HTU requests and the start of the element transfers result from the fact that the HTU can process only one transfer at a time.

Auto Switch Buffer Mode

If TMBA is set to auto switch mode, then the data stream will continue at the start of buffer B after all elements of buffer A have been transferred. This means that in the CPENA register, CP A is disabled and CP B is enabled automatically and buffer B uses its initial main memory address and initial frame counter to start. The same principle is valid for TMBB and buffer B.

The examples of Figure 24-7 assumes IETCOUNT=3 (Initial Element Transfer Count), IFTCOUNT=3 (Initial Frame Transfer Count, SIZE=0 (Size of Transfer = 32-bit) and ADDFM=0 (Addressing Mode Main Memory = Post Increment). So there are in total 9 32-bit values in buffer A and B. It also assumes IFADDRB=10h and IFADDRA=40h. "U" means uninitialized.

Figure 24-7. Timing Example for Auto Switch Buffer Mode



24.2.1.4 General Control Packet Behavior

The action defined by the selected mode will be performed at the end of the last frame, which has the frame counter value of 1. The one shot and auto switch mode will automatically update the CPENA register at this time. Note, that for all three modes listed above, it is possible to switch to the other buffer by writing to CPENA before the end of the current buffer is reached.

If a write access to CPENA happens while the last frame of DCP x (with frame counter = 1) is transferred then the priority is defined by [Table 24-1](#).

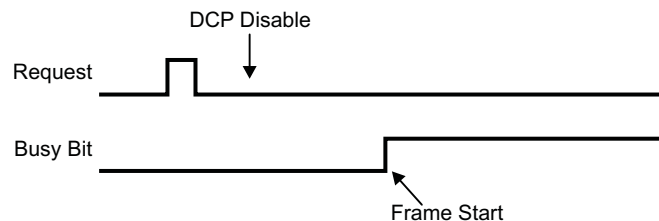
Table 24-1. CPENA / TMBx Priority Rules

Write access to CPENA bits ($2 \times x+1$) and ($2 \times x$) during the frame with frame counter = 1 ⁽¹⁾	Priority Rule
Disable: 01 --> 00 or 10 --> 00	Disabling the DCP by the write to CPENA has priority, TMBx is ignored.
Stay: 01 --> 01 or 10 --> 10	The write access to CPENA is ignored, TMBx has priority and defines the action.
Switch: 01 --> 10 or 10 --> 01	Switching the DCP by the write to CPENA has priority, TMBx is ignored.

⁽¹⁾ See read table of CPENA register ([Table 24-14](#))

There could be a case where the CPU wants to do main memory operations, but does not want the HTU modifying the main memory. It could happen that a request was already active, but the frame transfer hasn't started yet when the application disabled the control packets. The timing diagram in [Figure 24-8](#) shows this scenario.

Figure 24-8. Timing for Disabling Control Packets



Since the request for the transfer was already received before the DCPx is disabled, the HTU will still start the frame transfer. The application would poll the BUSYx bit during the time the DCPx was disabled and before the frame was started and would read a non-busy information. It then would start the main memory operations thinking all transfers have completed, however after some time the HTU will start the outstanding frame transfer and corrupt the main memory.

To avoid this, the application can set the VBUSHOLD bit to disable all transactions between the HTU and the main memory. It has to poll the BUSBUSY bit to ensure that no outstanding transactions on the bus are pending. The HTU will still receive all transfer requests from the N2HET, but it will not be able to transfer any data to or from the main memory, while the VBUSHOLD bit is set.

24.2.2 Arbitration of HTU Elements and Frames

- Frames do not interrupt each other. If a request occurs on DCP x while another frame runs on DCP y (and $x \neq y$), then the current frame completes before the new frame starts.
- If two or more request lines are active, the request line with the lower number (specified in the request number field of the corresponding N2HET instruction) is serviced first.

24.2.3 Conditions for Frame Transfer Interruption

If a frame is currently transferred on DCP x and one of the events listed below happens, then the event will (1.) clear the element counter of DCP x, (2.) stop new element transfers on DCP x (3.) clear the active busy bit of DCP x and (4.) disable DCP x in the CPENA register. The DCPs other than DCP x will not be affected.

- Request Lost Error of DCP x (with CORL bit set to 0).
- Parity Error of DCP x (with parity check enabled and COPE bit set to 0). See also [Section 24.2.6](#).
- Bus Error of DCP x.
- Memory Protection Error of DCP x (with memory protection enabled). See also [Section 24.2.5](#).
- Writing a 1 to a BUSY bit (belonging to DCP x) if that bit is 1. There is no effect if the BUSY bit is 0.
- Writing a 1 to the HTURES bit.

When a memory protection error occurs, the access to the protected address is blocked. The frame is stopped before the element, which caused the violation transfer, starts. All other errors will let the current element transfer finish.

In case of the Request Lost and Bus Error, one more element transfer goes on the bus, before the frame is actually stopped. Accordingly, the busy bit is cleared after the element, which follows the element that caused the error.

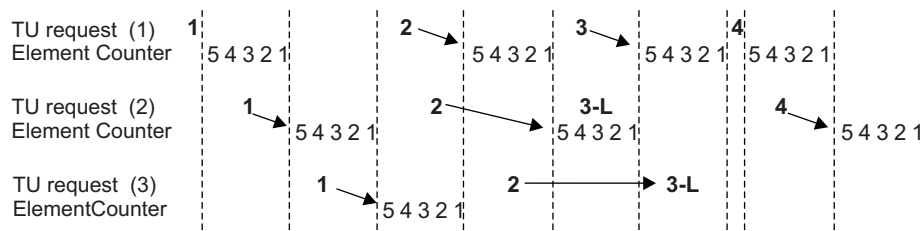
In case of the Bus Error, the counter for the element, which follows the element that caused the error, is captured to the ERRETC register field.

NOTE: If the HTUEN bit is cleared during a frame is transferred, then the frame will be completed before the HTU is disabled.

24.2.4 HTU Overload and Request Lost Detection

If the number of different HTU request sources is "high", the period between the requests is "short" and/or the initial element counter values are "big", then the HTU could get into a overload situation. In [Figure 24-9](#), all requests marked with "L" are lost, since their frame is not completed at the time the next request occurs. Each number in the rows "TU request (x)" represents a time, where the associated N2HET instruction generates a request on DCP x. The arrows in [Figure 24-9](#) point to the associated frame, which could be delayed compared to the request. The delays are caused by different frames, which are currently processed. The figure assumes that the CORL bit in the RLBECTRL register is set, which causes the DCP to stay enabled and let the data stream continue after a request lost error occurred on the DCP (see 3-L for TU request (2)).

Figure 24-9. Timing Example Including Lost Requests



Lost requests are signaled with the RLOSTFL register, and if enabled, can generate request lost interrupts.

If the CORL bit is set, a frame will be completed and the corresponding DCP stays enabled even if a request lost was generated during this frame.

In dual buffer mode, the request lost detection works continuously, independent of the CP switches.

24.2.4.1 Requests and Quiet Requests

In addition to generating too many transfer requests and thus overloading the HTU and not being able to transfer data at all, it can happen that inconsistent data is transferred. The following examples illustrate such scenarios.

In the examples below, the HTU reads a frame of three elements from the datafield of three different instructions. In Figure 24-10, the L3-Instruction generates the HTU request at time t2, t7, and so on. and the according frame (at t3). The frame is delayed because of the HTU load. However, as shown in Figure 24-10, the delay still allows the frame to complete before the datafield of instruction L1 is updated again. However, when the delay is longer (as shown in Figure 24-11), then the frame could fall into the N2HET loop (LRP), in which the N2HET updates the data fields of the L1, L2 and L3 instructions. In this case, the HTU could read inconsistent data as shown in the diagram. A wrong (new) value is read from L1 (at time t3), but correct ("old") values are read from L2 and L3 (at times t4 and t5).

Figure 24-10. Timing that Generates No Request Lost Error

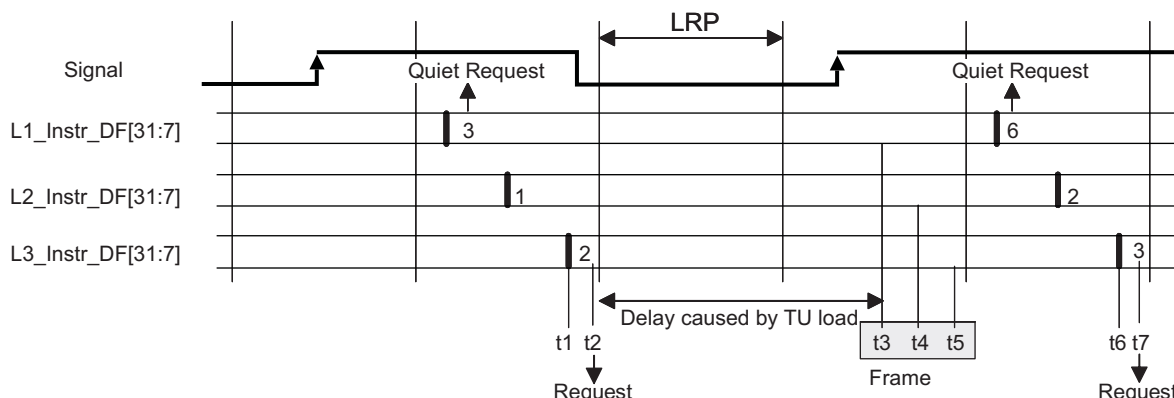
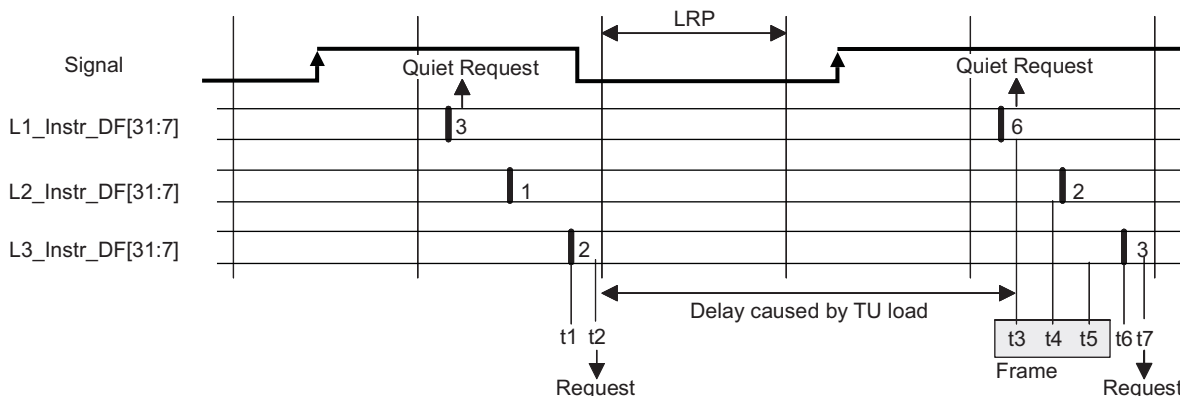


Figure 24-11. Timing that Generates a Request Lost Error



To prevent sending inconsistent data, the N2HET instructions are able to generate a **quiet request**, which does not originate a transfer but is only used by the HTU for consistency check. If a frame has not completed since the last request (or has not even started) at the time the quiet request occurs, then the HTU signals a request lost error. All instructions, which allow to generate a request can be configured to generate a quiet request instead. So in the examples of Figure 24-10 and Figure 24-11, instruction L1 should be configured to generate a quiet request and instruction L3 to generate a normal request. In the case of Figure 24-11, the corresponding bit in the RLOSTFL register will be set.

It is the responsibility of the N2HET software to enable a quiet request for the first instruction of an instruction block, which is addressed by DCP x, and to enable a normal request only for the last instruction of this block. Since enabling the quiet request should enable a proper request lost detection for DCP x, both N2HET instructions need to specify the same DCP x (reqnum=x).

The control fields of the HET instructions provide a 2-bit field to configure one of the following possibilities (as shown in Table 24-2). A 3-bit field in the program field will select which of the 8 Double Control Packets will be triggered by the request.

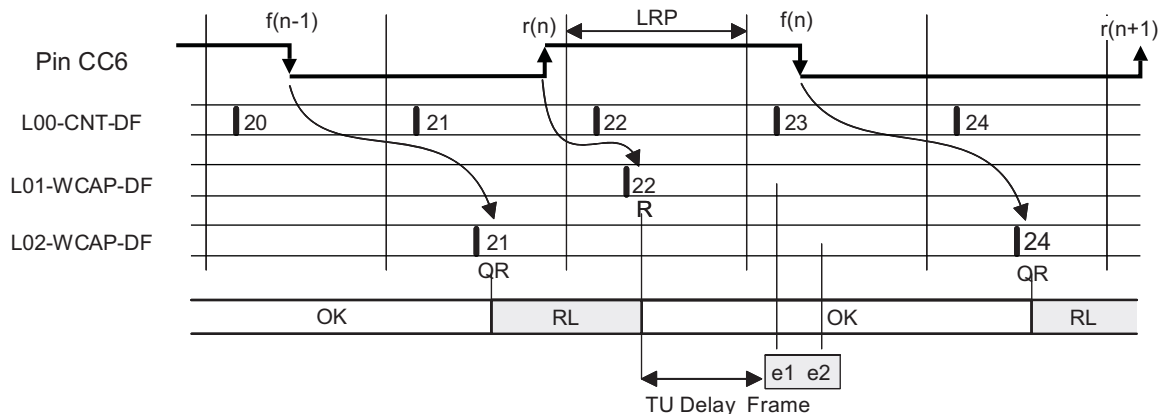
Table 24-2. Triggered Control Packets

Request Type Bit 1	Request Type Bit 0		Request Number
Don't care	0	No request	Specify number 0, 1,... or 7, which selects the HTU or DMA request line.
0	1	Generate normal request	
1	1	Generate quiet request	

In the case of very light HTU load, but higher signal requirements (for example, high frequency), the quiet request could also be used to define periods in which the data read by a control packet is safe. The following HET code will capture counter time stamps to the L1-WCAP data field after rising edges (at pin CC6) and to the L2-WCAP data field after falling edges (at pin CC6):

```
L0 CNT {reg=A, max=0x1FFFFFFF}
L1 WCAP {reqnum=3, request=GENREQ, event=RISE, reg=A, pin=CC6}
L2 WCAP {reqnum=3, request=QUIET, event=FALL, reg=A, pin=CC6}
; HET HRSHARE feature configured to assign both WCAPs to pin CC6
```

Figure 24-12. Timing Example for Two WCAP Instructions



The HTU frame will have two elements: The first gives the time stamp of the rising edge $r(n)$ and the second gives the time stamp of the previous falling edge $f(n-1)$. Using the code above, requests (R) and the quiet requests (QR) will occur at the times shown in Figure 24-12, and a request lost will only be signaled when the frame makes an access during the times marked with RL. So reading [22, 21] as frame elements is correct. If the signal frequency would increase, then a wrong pair [22, 23] could be read, but this will be signaled by a request lost error since at least $e2$ falls into the RL period.

24.2.5 Memory Protection

This feature allows restricting accesses to certain areas in memory in order to protect critical application data from unintentionally being manipulated by the HTU.

If the HTU memory protection feature is disabled, the full 4 GB address range can be accessed by the HTU without exception. There are two memory regions that start and end addresses can be configured.

With the HTU memory protection feature enabled, read and write accesses by the HTU IFADDRB and IFADDRB registers inside the defined regions are allowed. HTU access to its tightly-coupled memory is independent of the MPU, it routes through the dedicated HTU/N2HET bus using the IHADDR bits in the IHADDRCT register. See [Section 24.2](#) for details on the tightly-coupled bus. For accesses outside the regions, one of two modes is configurable:

- Any access performed by the HTU is forbidden and will be signaled to the ESM module. Write accesses will be blocked.
- Read access is allowed but write access will be blocked and signaled to the ESM module.

To use one region only, REG01ENA must be 0. Bits ACCR01, INTENA01, and register settings of MP1S and MP1E will be ignored.

To use both regions, the following rules must be followed:

1. Memory mapped region 0 covers a lower memory area as Memory mapped region 1.
2. REG01ENA is a 1 and REG0ENA is a 0.
3. ACCR01 is set for the desired access type, ACCR0 is ignored.
4. INTENA01 is set for the desired action, INTENA0 is ignored.

If an element transfer of DCP x generates a memory protection error, then:

1. The element counter of DCP x is cleared.
2. All new element transfers on DCP x are stopped.
3. The active busy bit of DCP x is cleared.
4. DCP x is disabled in the CPENA register. The DCPs other than DCP x will not be affected.
5. The FT flag will be set.
6. An error is signaled to the ESM module.

24.2.6 Control Packet RAM Parity Checking

The HTU module can detect parity errors in the DCP (Double Control Packet) RAM. DCP RAM parity checking is implemented using one parity bit per byte. Even or odd parity checking can be selected in the DEVCR1 register of the system module and can be enabled/disabled by a 4-bit key in the PCR register.

During a read access to the DCP RAM, the parity is calculated based on the data read from the RAM and compared with the good parity value stored in the parity bits. The parity check is performed when the HTU or any other master (for example, CPU) makes a read access to the DCP RAM. A read access within the RAM section of an initial or current DCP checks all 16 bytes of the DCP at a time (see also DCP memory map). For example, if a byte read access happens for DCP RAM address 0, but there is a parity error at byte address Ch then the parity error will occur and the captured parity address will be Ch and not 0. The address of the byte in which the error occurred can be read from the PAR register. If successive DCP RAM read accesses generate multiple parity errors, only the address of the first detected error will be captured and the PAR register will not be updated by subsequent errors until it is read by the application. When multiple errors in a 16 byte word are detected, only the address of the lowest byte will be captured.

The application can decide whether to stop any transfers when a parity error is detected or to continue transferring data. If the COPE (Continue On Parity Error) bit is 0 and parity checking is enabled, then the HTU will not start the frame and the corresponding DCP will be automatically disabled in the CPENA register. If a master other than the HTU (for example, CPU) reads the RAM section of DCP x and a parity error is detected during this read access, while the parity check is enabled and the COPE bit is 0, then the DCP x will be automatically disabled in the CPENA register. If a frame for this DCP x is ongoing during

this read access, then in addition the element counter of DCP x is cleared, all new element transfers on DCP x are stopped and the active busy bit of DCP x is cleared. With COPE set to 1 and the parity check enabled, the parity checking will still be performed, but the data transfer of an active DCP continues after a parity error was detected for this DCP. So neither the DCP with the parity error will be disabled nor the frame will be stopped.

After a DCP is enabled (with CPENA using BIM=0), then at the start of the first frame, the HTU performs the parity check only on the initial DCP, since it does not need the current DCP information. For further frames, the HTU performs the parity check for both initial and current DCP, since it needs both information.

On a parity error detection, an error will also be signaled to the ESM module.

24.2.6.1 Parity Bit Mapping and Testing

To test the parity checking mechanism, the parity RAM can be made accessible in order to allow manual fault insertion. Once the TEST bit is set, the parity bits are mapped to address FF4E 0200h.

When in test mode (the parity RAM is accessible), no parity checking will be done when reading from parity RAM, but parity checking will still be performed for read accesses to the DCP RAM.

Table 24-3 and Table 24-4 show how the corresponding parity bits of the DCP RAM bytes are mapped into the memory.

Table 24-3. DCP RAM

Bit	31	24	23	16	15	8	7	0
FF4E 0000h	Byte 0		Byte 1		Byte 2		Byte 3	
FF4E 0004h	Byte 4		Byte 5		Byte 6		Byte 7	
FF4E 0008h	Byte 8		Byte 9		Byte 10		Byte 11	
FF4E 000Ch	Byte 12		Byte 13		Byte 14		Byte 15	

Table 24-4. DCP Parity RAM

Bit	24	16	8	0
FF4E 0200h	P0	P1	P2	P3
FF4E 0204h	P4	P5	P6	P7
FF4E 0208h	P8	P9	P10	P11
FF4E 020Ch	P12	P13	P14	P15

Each byte in DCP RAM has its own parity bit in the DCP Parity RAM. P0 is the parity bit for byte 0, P1 is the parity bit for byte 1, and so on.

24.2.6.2 Initializing Parity Bits

After device power up, the DCP RAM content including the parity bit cannot be guaranteed. In order to avoid parity failures, when reading DCP RAM, the RAM has to be initialized first. This can simply be done by writing known values into the RAM by software and the corresponding parity bit will be automatically calculated.

Another possibility to initialize the DCP memory and its parity bits is to use the system module, which is an on-chip module external to the HTU. This module can start the automatic initialization of all RAMs on the microcontroller including the HTU DCP RAM. This function initializes the complete DCP RAM to 0 when activated by the system module. Depending on the even/odd parity selection, all parity bits will be calculated accordingly. The HTUEN bit must be cleared and the parity functionality must be enabled (by PARITY_ENA) during the automatic DCP RAM initialization. If HTUEN is 1 when the initialization is triggered by the system module, then the initialization will not be performed and the HTU operation is not affected. If a 1 is written to HTUEN during the initialization, then the HTUEN bit will be set but the HTU will not be enabled before the initialization completes.

24.3 Use Cases

24.3.1 Example: Single Element Transfer with One Trigger Request

This example considers the case that the HTU fills a RAM buffer in the main (CPU) data RAM. The HTU reads from the instruction which generates the HTU requests.

This example uses a PCNT instruction. Every time the PCNT has captured a new pulse or period value, it will automatically generate a transfer request to the HTU, which then transfers the value from the N2HET RAM to the buffer RAM. So over time consecutive locations in the RAM buffer can be filled with consecutive measurement values captured into the N2HET RAM data field of the same PCNT instruction without loading or interrupting the CPU.

24.3.2 Example: Multiple Element Transfer with One Trigger Request

The following example shows how the HTU could be used to fill a RAM buffer with a data stream including different types of measurement values belonging to the same N2HET input signal (on one pin): Time stamp values (WCAP), edge counter values (ECNT) and last period values (PCNT).

Figure 24-13 shows the timing and Table 24-5 shows the byte addresses of the program- (PF), control- (CF), data- (DF) and reserved field (res) of the WCAP-ECNT-PCNT instruction block. The timing and code example assumes that all three instructions are assigned to the same N2HET pin.

Figure 24-13. Timing of the WCAP, ECNT, PCNT Example

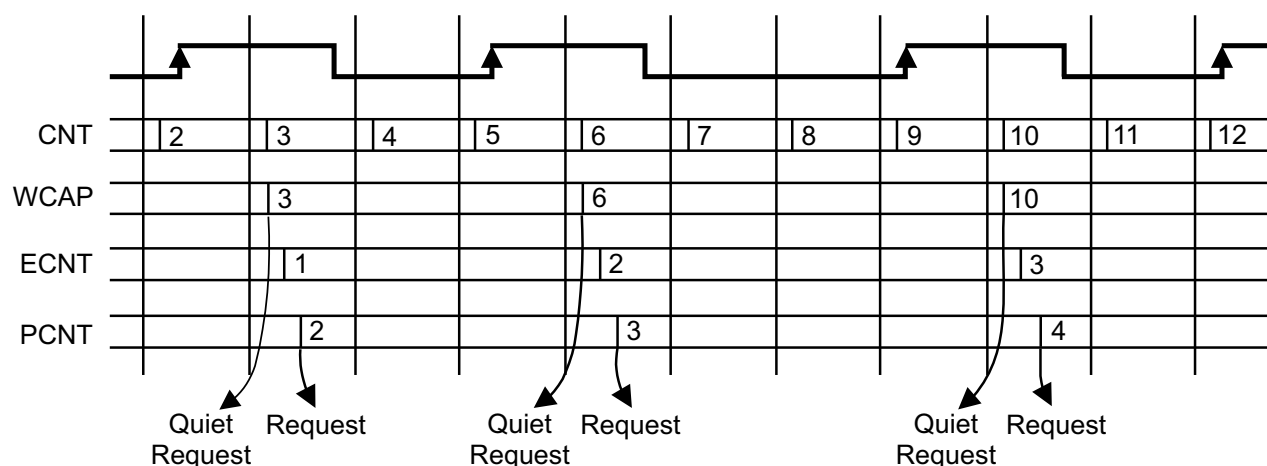


Table 24-5. Field Addresses of the WCAP, ECNT, PCNT Example

	PF	CF	DF	Res
WCAP	30h	34h	38h	3Ch
ECNT	40h	44h	48h	4Ch
PCNT	50h	54h	58h	5Ch

In the HET code the HTU request is enabled only for the last instruction (PCNT) of the WCAP-ECNT-PCNT block. When the PCNT condition is true, it will cause the generated HTU frame to perform three HTU element reads from the data fields of WCAP, ECNT, and PCNT.

32-Bit-Transfer of data fields:

[Table 24-6](#) shows how the internal element counter, frame counter and the address registers change over time for the example described above. Every time the PCNT instruction captures a new value it generates a request to the HTU, which starts a frame. At the end of each frame the frame counter decrements.

Table 24-6. 32-Bit-Transfer of Data Fields⁽¹⁾

Frame Counter	3			2			1		
Element Counter	3	2	1	3	2	1	3	2	1
Source Address (HET)	38h	48h	58h	38h	48h	58h	38h	48h	58h
Destination Address (main CPU RAM)	70h	74h	78h	7Ch	80h	84h	88h	8Ch	90h

⁽¹⁾ Shows the byte addresses

The destination buffer is filled with the WCAP, ECNT, and PCNT data field values as shown in [Table 24-7](#).

Table 24-7. Destination Buffer Values

Address	Frame Count	Instruction	Value
70h	3	WCAP	3
74h	3	ECNT	1
78h	3	PCNT	2
7Ch	2	WCAP	6
80h	2	ECNT	2
84h	2	PCNT	3
88h	1	WCAP	10
8Ch	1	ECNT	3
90h	1	PCNT	4

The corresponding setup of the HTU control packet for this example is as follows:

```

IHADDR    = 0x38           // points to WCAP data field
IFADDRA   = 0x70           // points to buffer
ITCOUNT  [frame count = 3] [element count = 3]
IHADDRCT  = [DIR: Read HET and write to full address]
            [SIZE: 32 bit]
            [ADDMH: Increment HET address by 16 bytes]
            [ADDMP: Post increment full address mode]
            [Any transfer mode]

```


24.3.3 Example: 64-Bit-Transfer of Control Field and Data Fields

Table 24-8 shows how the internal element counter, frame counter and the address registers change over time assuming the same example as in Section 24.3.2, but now with a transfer size set to 64-bit. The HET address now points to the control field of the instruction, so CF and DF are transferred as 64 bit data.

Table 24-8. 64-Bit-Transfer of Control Field and Data Fields⁽¹⁾

Frame Counter	3			2			1		
Element Counter	3	2	1	3	2	1	3	2	1
HET (Source) Address	34h	44h	54h	34h	44h	54h	34h	44h	54h
Full (Destination) Address	70h	78h	80h	88h	90h	98h	A0h	A8h	B0h

⁽¹⁾ Shows the byte addresses.

The destination buffer is filled with the WCAP, ECNT, and PCNT control and data field values as shown on the right in Table 24-9.

Table 24-9. Destination Buffer Values

Address	Frame Count	Instruction	Value
70h	3	WCAP	Control Field Value
74h	3	WCAP	3
78h	3	ECNT	Control Field Value
7Ch	3	ECNT	1
80h	3	PCNT	Control Field Value
84h	3	PCNT	2
88h	2	WCAP	Control Field Value
8Ch	2	WCAP	6
90h	2	ECNT	Control Field Value
94h	2	ECNT	2
98h	2	PCNT	Control Field Value
9Ch	2	PCNT	3
A0h	1	WCAP	Control Field Value
A4h	1	WCAP	10
A8h	1	ECNT	Control Field Value
ACh	1	ECNT	3
B0h	1	PCNT	Control Field Value
B4h	1	PCNT	4

The necessary setup of the HTU control packet (see Section 24.5) for this example is as follows:

```

IHADDR = 0x34 (points to WCAP control field)
IFADDR = 0x70 (points to buffer)
ITCOUNT [frame count = 3] [element count = 3]
IHADDRCT = [DIR: Read HET and write to full address]
           [SIZE: 64 bit]
           [ADDMH: Increment HET address by 16 bytes]
           [ADDMF: post increment full address mode]
           [Any transfer mode]

```

For different applications, which have the transfer direction set for reading the buffer and writing to HET fields, the 64-bit transfer could be used to change the conditional addresses together with a new data field.

24.4 HTU Control Registers

[Table 24-10](#) provides a summary of the registers. The registers support 8-bit, 16-bit, and 32-bit writes. The offset is relative to the associated peripheral select. See the following sections for detailed descriptions of the registers. The base address for the control registers is FFF7 A400h for HTU1 and FFF7 A500h for HTU2. The address locations not listed, are reserved.

Table 24-10. HTU Control Registers

Offset	Acronym	Register Description	Section
00h	HTU GC	Global Control Register	Section 24.4.1
04h	HTU CPENA	Control Packet Enable Register	Section 24.4.2
08h	HTU BUSY0	Control Packet Busy Register 0	Section 24.4.3
0Ch	HTU BUSY1	Control Packet Busy Register 1	Section 24.4.4
10h	HTU BUSY2	Control Packet Busy Register 2	Section 24.4.5
14h	HTU BUSY3	Control Packet Busy Register 3	Section 24.4.6
18h	HTU ACPE	Active Control Packet and Error Register	Section 24.4.7
20h	HTU RLBECTRL	Request Lost and Bus Error Control Register	Section 24.4.8
24h	HTU BFINTS	Buffer Full Interrupt Enable Set Register	Section 24.4.9
28h	HTU BFINTC	Buffer Full Interrupt Enable Clear Register	Section 24.4.10
2Ch	HTU INTMAP	Interrupt Mapping Register	Section 24.4.11
34h	HTU INTOFF0	Interrupt Offset Register 0	Section 24.4.12
38h	HTU INTOFF1	Interrupt Offset Register 1	Section 24.4.13
3Ch	HTU BIM	Buffer Initialization Mode Register	Section 24.4.14
40h	HTU RLOSTFL	Request Lost Flag Register	Section 24.4.15
44h	HTU BFINTFL	Buffer Full Interrupt Flag Register	Section 24.4.16
48h	HTU BERINTFL	BER Interrupt Flag Register	Section 24.4.17
4Ch	HTU MP1S	Memory Protection 1 Start Address Register	Section 24.4.18
50h	HTU MP1E	Memory Protection 1 End Address Register	Section 24.4.19
54h	HTU DCTRL	Debug Control Register	Section 24.4.20
58h	HTU WPR	Watch Point Register	Section 24.4.21
5Ch	HTU WMR	Watch Mask Register	Section 24.4.22
60h	HTU ID	Module Identification Register	Section 24.4.23
64h	HTU PCR	Parity Control Register	Section 24.4.24
68h	HTU PAR	Parity Address Register	Section 24.4.25
70h	HTU MPCS	Memory Protection Control and Status Register	Section 24.4.26
74h	HTU MP0S	Memory Protection 0 Start Address Register	Section 24.4.27
78h	HTU MP0E	Memory Protection 0 End Address Register	Section 24.4.28

24.4.1 Global Control Register (HTU GC)

Figure 24-14. Global Control Register (HTU GC) [offset = 00]

31	25	24	23	17	16
Reserved		VBUSHOLD		Reserved	HTUEN
R-0		R/WP-0		R-0	R/WP-0
15	9	8	7	1	0
Reserved		DEBM		Reserved	HTURES
R-0		R/WP-0		R-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

Table 24-11. Global Control Register (HTU GC) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	VBUSHOLD	0 1	Hold the VBUS bus The VBUS is not held. The VBUSHOLD bit holds the bus used to transfer data between the HTU and the N2HET module. When the BUS_BUSY bit is 0 then the bus is no longer busy. While the bus is held, requests will still be accepted. They will be acted upon when the VBUSHOLD is 0. Request lost conditions will be detected and interrupts generated if they are enabled.
23-17	Reserved	0	Reads return 0. Writes have no effect.
16	HTUEN	0 1	Transfer Unit Enable Bit The Transfer Unit is disabled. The Transfer Unit is enabled. The configuration registers and control packets should be set up first before the HTUEN bit is set to 1 to prevent it from carrying out unintended bus transactions. If the HTUEN bit is cleared to 0 during a frame is transferred, then the frame will be completed before the HTU is disabled. The HTUEN bit must be cleared to 0 and the parity functionality must be enabled (by PARITY_ENA) during the automatic DCP RAM initialization (see Initializing Parity Bits). If HTUEN is 1 when the initialization is triggered by the system module, then the initialization will not be performed and the HTU operation is not affected. If a 1 is written to HTUEN during the initialization, then the HTUEN bit will be set but the HTU will not be enabled before the initialization completes. Note: If HTU is disabled during a frame transfer, then the ongoing current frame will be completed before the HTU module is disabled. If enabled again, then the transfer will restart from the initial frame count for the CP programmed.
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	DEBM	0 1	Debug Mode The Transfer Unit is stopped in debug mode. The HTU will complete the current frame, but not start any new frames. It will also ignore all requests from the HET and not generate any request lost signals. The Transfer Unit continues operation in debug mode. Note: Since the HET has also an "ignore suspend" bit, there a several possibilities for the behavior of the HET and HTU in suspend mode.
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	HTURES	0 1	HTU Software Reset Request Reset request is not issued to the HTU module. Writing a 0 has no effect. Reset request is issued to the HTU module. Ongoing element transfers will be completed, before resetting the complete HTU module, similar to a hardware reset. The HTURES bit will also be cleared. The recommended order of operations is: - Set the software reset bit. This also clears HTUEN. - Wait for the HTURES bit to clear. - Configure the HTU registers and packets. - Set the HTUEN bit to begin operation.

24.4.2 Control Packet Enable Register (HTU CPENA)

This register enables or disables the individual double control packets (DCP).

Figure 24-15. Control Packet Enable Register (HTU CPENA) [offset = 04h]

31	Reserved	16
	R-0	
15	CPENA	0
	R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

Table 24-12. Control Packet Enable Register (HTU CPENA) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	CPENA		CP Enable Bits Bits (2*x) and (2*x+1) of CPENA control the double control packet (DCP) x (whereby x must be within 0,1,...,7). See Table 24-13 for write rules. See Table 24-14 for read rules.

Table 24-13. CPENA Write Results

Bit (2*x+1)	Bit (2*x)	Control packets (CP) B and A of DCP x are affected as follows:
0	0	CP B and A are not changed.
0	1	CP B is disabled and CP A are enabled simultaneously.
1	0	CP B is enabled and CP A are disabled simultaneously.
1	1	CP B and CP A are both disabled simultaneously.

Table 24-14. CPENA Read Results

Bit (2*x+1)	Bit (2*x)	State of DCP:
0	0	The DCP is disabled.
0	1	CP B is disabled and CP A is enabled.
1	0	CP B is enabled and CP A is disabled.
1	1	Cannot be read.

- The conditions listed in [Section 24.2.3](#) can automatically disable DCP x. In this case, bits (2*x) and (2*x+1) are both automatically set to 0.
- When bits (2*x) and (2*x+1) change from 00 to 01 or from 00 to 10 caused by a write access to CPENA, then old pending requests on the corresponding request line are cleared. This means only new requests which occur **after** this write access cause the first HTU transfer for this DCP. This is **not** the case when **switching** CPs (from 10 to 01 or from 01 to 10).
- CP A and/or CP B of a DCP can be configured to one-shot, circular or auto-switch transfer mode via the TMBA or TMBB bits in the IHADDRCT control packet configuration. If a write access to CPENA occurs during the last frame of a buffer (with frame counter = 1) then the action defined by the write access to CPENA and the action defined by TMBx can contradict. The priority rules for this case are given in [Table 24-1](#).

24.4.3 Control Packet (CP) Busy Register 0 (HTU BUSY0)

This register displays the status of individual control packets.

Figure 24-16. Control Packet (CP) Busy Register 0 (HTU BUSY0) [offset = 08h]

31	25	24	23	17	16
Reserved	BUSY0A	Reserved	BUSY0B		
R-0	R/W1CP-0	R-0	R/W1CP-0		
15	9	8	7	1	0
Reserved	BUSY1A	Reserved	BUSY1B		
R-0	R/W1CP-0	R-0	R/W1CP-0		

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode only to clear the bit; -n = value after reset

Table 24-15. Control Packet (CP) Busy Register 0 (HTU BUSY0) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	BUSY0A		Busy Flag for CP A of DCP 0
23-17	Reserved	0	Reads return 0. Writes have no effect.
16	BUSY0B		Busy Flag for CP B of DCP 0
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	BUSY1A		Busy Flag for CP A of DCP 1
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	BUSY1B		Busy Flag for CP B of DCP 1

The bit is **set** when the frame on the according control packet **starts** (as shown in the diagram below, there could be a delay between the request and the start of the frame).

The bit is automatically **cleared** at any of the following conditions:

- At the end of a frame.
- Writing a 1 to a BUSY bit (of DCP x) if that bit is 1. This will:
 - clear the element counter of DCP x
 - stop all new element transfers on DCP x
 - clear the busy bit
 - and disable DCP x in the CPENA register.

There is no effect, if the BUSY bit is 0.

- At the conditions listed in [Section 24.2.3](#).

A write access to the CPENA register can stop a control packet (CP) in single buffer mode or it can switch to the other CP of a DCP in dual buffer mode. If stopping or switching occurs while a frame runs on the currently active control packet, the CPU can poll the busy bit to determine when it is safe to read the buffer.

24.4.4 Control Packet (CP) Busy Register 1 (HTU BUSY1)

This register displays the status of individual control packets.

Figure 24-17. Control Packet (CP) Busy Register 1 (HTU BUSY1) [offset = 0Ch]

31	25	24	23	17	16
Reserved	BUSY2A	Reserved	BUSY2B		
R-0	R/W1CP-0	R-0	R/W1CP-0		
15	9	8	7	1	0
Reserved	BUSY3A	Reserved	BUSY3B		
R-0	R/W1CP-0	R-0	R/W1CP-0		

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode only to clear the bit; -n = value after reset

Table 24-16. Control Packet (CP) Busy Register 1 (HTU BUSY1) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	BUSY2A		Busy Flag for CP A of DCP 2
23-17	Reserved	0	Reads return 0. Writes have no effect.
16	BUSY2B		Busy Flag for CP B of DCP 2
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	BUSY3A		Busy Flag for CP A of DCP 3
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	BUSY3B		Busy Flag for CP B of DCP 3

See [Section 24.4.3](#) for more details.

24.4.5 Control Packet (CP) Busy Register 2 (HTU BUSY2)

Figure 24-18. Control Packet (CP) Busy Register 2 (HTU BUSY2) [offset = 10h]

31	25	24	23	17	16
Reserved	BUSY4A	Reserved	BUSY4B		
R-0	R/W1CP-0	R-0	R/W1CP-0		
15	9	8	7	1	0
Reserved	BUSY5A	Reserved	BUSY5B		
R-0	R/W1CP-0	R-0	R/W1CP-0		

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode only to clear the bit; -n = value after reset

Table 24-17. Control Packet (CP) Busy Register 2 (HTU BUSY2) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	BUSY4A		Busy Flag for CP A of DCP 4
23-17	Reserved	0	Reads return 0. Writes have no effect.
16	BUSY4B		Busy Flag for CP B of DCP 4
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	BUSY5A		Busy Flag for CP A of DCP 5
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	BUSY5B		Busy Flag for CP B of DCP 5

24.4.6 Control Packet (CP) Busy Register 3 (HTU BUSY3)

Figure 24-19. Control Packet (CP) Busy Register 3 (HTU BUSY3) [offset = 14h]

31	25	24	23	17	16
Reserved	BUSY6A	Reserved	BUSY6B		
R-0	R/W1CP-0	R-0	R/W1CP-0		
15	9	8	7	1	0
Reserved	BUSY7A	Reserved	BUSY7B		
R-0	R/W1CP-0	R-0	R/W1CP-0		

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode only to clear the bit; -n = value after reset

Table 24-18. Control Packet (CP) Busy Register 3 (HTU BUSY3) Field Descriptions

Bit	Field	Value	Description
31-25	Reserved	0	Reads return 0. Writes have no effect.
24	BUSY6A		Busy Flag for CP A of DCP 6
23-17	Reserved	0	Reads return 0. Writes have no effect.
16	BUSY6B		Busy Flag for CP B of DCP 6
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	BUSY7A		Busy Flag for CP A of DCP 7
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	BUSY7B		Busy Flag for CP B of DCP 7

24.4.7 Active Control Packet and Error Register (HTU ACPE)

Figure 24-20. Active Control Packet and Error Register (HTU ACPE) [offset = 18h]

31	30	29	28	24	23	20	19	16
ERRF	Reserved	ERRETC	Reserved	ERRCPN				
R/W1CP-0	R-0	R-0	R-0	R-0	R-0			
15	14	13	12	8	7	4	3	0
TIPF	BUSBUSY	Rsvd	CETCOUNT	Reserved	NACP			
R-0	R-0	R-0	R-0	R-0	R-0			

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode only to clear the bit; -n = value after reset

Table 24-19. Active Control Packet and Error Register (HTU ACPE) Field Descriptions

Bit	Field	Value	Description
31	ERRF	0 1	Error Flag No error occurred. This bit is set when one of the conditions listed at ERRETC is fulfilled and ERRETC and ERRCPN are captured. Once ERRF is set, it is cleared when reading the upper 16-bit word of the ACPE register or the complete 32-bit register. It is also cleared when writing a 1 to ERRF. ERRF can be used to indicate if ERRETC and ERRCPN contain new unread data.
30-29	Reserved	0	Reads return 0. Writes have no effect.

Table 24-19. Active Control Packet and Error Register (HTU ACPE) Field Descriptions (continued)

Bit	Field	Value	Description						
28-24	ERRETC		Error Element Transfer Count If one of the following conditions happens the current element transfer counter of the control packet (specified by ERRCPN) is captured to ERRETC. Please see Section 24.2.3. - Request Lost Error of control packet specified by ERRCPN. This is independent of the CORL bit. - Parity Error of control packet specified by ERRCPN. This requires the parity check to be enabled, but is independent of the COPE bit. - Bus Error of control packet specified by ERRCPN. - Memory Protection Error of control packet specified by ERRCPN. This requires the memory protection to be enabled. - Writing a 1 to a BUSY bit, which belongs to the control packet specified by ERRCPN, if that bit is 1. There is no effect, if the BUSY bit is 0. ERRETC is frozen from being updated until the upper 16-bit word of the ACPE register or the complete 32-bit register is read by the CPU. After this read, the HTU will update ERRETC if one of the above conditions is fulfilled again. During debugging, ERRETC stays frozen even when reading the upper 16-bit word or the 32-bit register.						
23-20	Reserved	0	Reads return 0. Writes have no effect.						
19-16	ERRCPN		Error Control Packet Number If one of the conditions listed at ERRETC happens the number of the control packet, which caused the condition, is captured to ERRCPN. <table><thead><tr><th>Control Packet</th><th>ERRCPN Value</th></tr></thead><tbody><tr><td>CP A of DCP x</td><td>2 x</td></tr><tr><td>CP B of DCP x</td><td>2 x+1</td></tr></tbody></table> With x = 0,1,...or 7 ERRCPN is frozen from being updated until the upper 16-bit word of the ACPE register or the complete 32-bit register is read by the CPU. After this read, the HTU will update ERRCPN if one of the above conditions is fulfilled again. During debugging, ERRCPN stays frozen even when reading the upper 16-bit word or the 32-bit register. If one of the conditions is fulfilled, ERRETC and ERRCPN are updated simultaneously.	Control Packet	ERRCPN Value	CP A of DCP x	2 x	CP B of DCP x	2 x+1
Control Packet	ERRCPN Value								
CP A of DCP x	2 x								
CP B of DCP x	2 x+1								
15	TIPF	0 1	Transfer in Progress Flag 0 1 No transfers are in progress. A transfer is currently active. This bit is the result of a logical OR function of all BUSYxx flags of the 4 BUSYx registers.						
14	BUSBUSY	0 1	Bus is Busy 0 1 Bus between N2HET and HTU is not busy. When BUSBUSY is 1, the bus is busy with a transfer. It is different from TIPF above because BUSBUSY will go low after VBUSHOLD is set to 1 and no transfers are pending between the HTU and the main memory. TIPF will remain 1, if a transfer is still pending and VBUSPHOLD is 1.						
13	Reserved	0	Reads return 0. Writes have no effect.						
12-8	CETCOUNT		Current Element Transfer Count CETCOUNT shows the current element transfer counter for the frame that is currently processed. If the HTU does not currently transfer any frame, CETCOUNT is 0. CETCOUNT is updated after the write part of a transfer. There is a period of up to 7 cycles between the time the CETCOUNT is 0 and the HTU is finished updating the current DCP (and the CPENA registers, if the required conditions are fulfilled).						
7-4	Reserved	0	Reads return 0. Writes have no effect.						
3-0	NACP		Number of Active Control Packet Indicates which CP currently processes a frame. <table><thead><tr><th>Active or Recent DCP</th><th>NACP Value</th></tr></thead><tbody><tr><td>CP A of DCP x</td><td>2 x</td></tr><tr><td>CP B of DCP x</td><td>2 x+1</td></tr></tbody></table> With x = 0,1,...or 7 NACP is updated at the time the frame starts on the according CP, and it is updated with a new value when a frame starts on a different CP. Note, that there can be a delay between the request and the start of the frame.	Active or Recent DCP	NACP Value	CP A of DCP x	2 x	CP B of DCP x	2 x+1
Active or Recent DCP	NACP Value								
CP A of DCP x	2 x								
CP B of DCP x	2 x+1								

24.4.8 Request Lost and Bus Error Control Register (HTU RLBECTRL)

Figure 24-21. Request Lost and Bus Error Control Register (HTU RLBECTRL) [offset = 20h]

31					17		16
Reserved						BERINTENA	
R-0						R/WP-0	
15		9	8	7		1	0
Reserved			CORL	Reserved		RLINTENA	
R-0			R/WP-0	R-0		R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

Table 24-20. Request Lost and Bus Error Control Register (HTU RLBECTRL) Field Descriptions

Bit	Field	Value	Description
31-17	Reserved	0	Reads return 0. Writes have no effect.
16	BERINTENA	0	Bus Error Interrupt Enable Bit The bus error interrupt is disabled for all DCPs.
		1	The bus error interrupt is enabled for all DCPs.
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	CORL	0	Continue On Request Lost Error Stop current frame on request lost detection. Please see Section 24.2.3 .
		1	If CORL is 1 and DCP x is enabled, then DCP x will stay enabled after a request lost condition on DCP x and element transfers will continue.
7-1	Reserved	0	Reads return 0. Writes have no effect.
0	RLINTENA	0	Request Lost Interrupt Enable Bit The request lost interrupt is disabled for all DCPs. Disabling RLINTENA will not clear the flags in the RLOSTFL register.
		1	The request lost interrupt is enabled for all DCPs. If bits are set in the RLOSTFL flag register at the time RLINTENA is (re-) enabled, then the according interrupt(s) will occur (in the order of the priority of the request lines).

24.4.9 Buffer Full Interrupt Enable Set Register (HTU BFINTS)

This registers allows to enable the buffer full interrupts for the different control packets. Reading registers BFINTS and BFINTC will return the same bits indicating the status which interrupt is enabled (1) or disabled (0).

Figure 24-22. Buffer Full Interrupt Enable Set Register (HTU BFINTS) [offset = 24h]

31	Reserved	16
R-0		
15	BFINTENA	0
R/WP-0		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

Table 24-21. Buffer Full Interrupt Enable Set Register (HTU BFINTS) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	BFINTENA	0	Interrupt is disabled. Writing a 0 has no effect.
		1	Writing to bit (2*x) enables the interrupt for CP A of DCP x. Writing to bit (2*x+1) enables the interrupt for CP B of DCP x.

24.4.10 Buffer Full Interrupt Enable Clear Register (HTU BFINTC)

This registers allows to disable the buffer full interrupts for the different control packets. Reading registers BFINTS and BFINTC will return the same bits indicating the status which interrupt is enabled (1) or disabled (0)

Figure 24-23. Buffer Full Interrupt Enable Clear Register (HTU BFINTC) [offset = 28h]

31	Reserved	16
R-0		
15	BFINTDIS	0
R/WP-0		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

Table 24-22. Buffer Full Interrupt Enable Clear Register (HTU BFINTC) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	BFINTDIS	0	Interrupt is disabled. Writing a 0 has no effect.
		1	Writing to bit (2*x) disables the interrupt for CP A of DCP x. Writing to bit (2*x+1) disables the interrupt for CP B of DCP x.

24.4.11 Interrupt Mapping Register (HTU INTMAP)

Figure 24-24. Interrupt Mapping Register (HTU INTMAP) [offset = 2Ch]

31	Reserved	17	16
	R-0		MAPSEL
			R/WP-0
15	CPINTMAP		0
	R/WP-0		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

Table 24-23. Interrupt Mapping Register (HTU INTMAP) Field Descriptions

Bit	Field	Value	Description
31-17	Reserved	0	Reads return 0. Writes have no effect.
16	MAPSEL	0	Interrupt Mapping Select Bit If MAPSEL is 0, then one bit of CPINTMAP selects one of two interrupt priorities 0 or 1 for the buffer full interrupt for the according CP. The request lost and bus error interrupts of all CPs are set to priority 0, independent of CPINTMAP.
		1	If MAPSEL is 1, then one bit of CPINTMAP determines if the buffer full , request lost and bus error interrupts of the according CP are assigned either to interrupt line 0 or to 1.
15-0	CPINTMAP	0	CP Interrupt Mapping Bits Interrupt of CP A (bit 2-x) of DCP x is mapped to interrupt line 0. Interrupt of CP B (bit 2*x+1) of DCP x is mapped to interrupt line 0.
		1	Interrupt of CP A (bit 2-x) of DCP x is mapped to interrupt line 1. Interrupt of CP B (bit 2*x+1) of DCP x is mapped to interrupt line 1.

24.4.12 Interrupt Offset Register 0 (HTU INTOFF0)

The INTOFF0 register reflects the highest priority interrupt flag bit set in the BERINTFL, RLOSTFL, or BFINTFL flag registers with the appropriate CPINTMAP bit set to 0. The priority order (from high to low) is: BER, RLOST, buffer-full. Interrupts for request lines with lower number have higher priority.

Figure 24-25. Interrupt Offset Register 0 (HTU INTOFF0) [offset = 34h]

31	Reserved															16	
R-0																	
15	Reserved					10	9	8	7	Reserved			4	3	CPOFF0		0
R-0					R-0			R-0			R-0						

LEGEND: R = Read only; -n = value after reset

Table 24-24. Interrupt Offset Register 0 (HTU INTOFF0) Field Descriptions

Bit	Field	Value	Description
31-10	Reserved	0	Reads return 0. Writes have no effect.
9-8	INTTYPE0	0 1h 2h 3h	Interrupt Type of Interrupt Line 0. Indicates whether a buffer-full, RLOST, or BER interrupt, assigned to interrupt line 0, is currently pending. No interrupt. Interrupt caused by full buffer on CP/DCP specified by CPOFF0. RLOST interrupt generated by CP/DCP specified by CPOFF0. BER interrupt generated by CP/DCP specified by bits CPOFF0.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	CPOFF0	0 1h 2h 3h 4h 5h 6h 7h 8h 9h Ah Bh Ch Dh Eh Fh	CP Offset. Indicates for which control packet the interrupt is pending, which is classified by INTTYPE0 and is assigned to interrupt line 0. DCP 0, CP A DCP 0, CP B DCP 1, CP A DCP 1, CP B DCP 2, CP A DCP 2, CP B DCP 3, CP A DCP 3, CP B DCP 4, CP A DCP 4, CP B DCP 5, CP A DCP 5, CP B DCP 6, CP A DCP 6, CP B DCP 7, CP A DCP 7, CP B

NOTE: Reading CPOFF0 will clear the bit generating the current interrupt from appropriate flag register (BERINTFL, RLOSTFL, or BFINTFL), except when in debug mode where reading CPOFF0 will have no effect on the flag registers.

In order to read INTTYPE0 and CPOFF0 simultaneously, always read this register using word or half-word but not using byte accesses.

24.4.13 Interrupt Offset Register 1 (HTU INTOFF1)

This register is organized identically to the INTOFF0 register. The difference is that INTOFF1 reflects the highest priority interrupt flag bit set in the BERINTFL, RLOSTFL, or BFINTFL flag registers with the appropriate CPINTMAP bit set to 1.

Figure 24-26. Interrupt Offset Register 1 (HTU INTOFF1) [offset = 38h]

31	Reserved																16
R-0																	
15	Reserved				10	9	8	7	Reserved				4	3	CPOFF1		0
R-0					R-0			R-0					R-0				

LEGEND: R = Read only; -n = value after reset

Table 24-25. Interrupt Offset Register 1 (HTU INTOFF1) Field Descriptions

Bit	Field	Value	Description
31-10	Reserved	0	Reads return 0. Writes have no effect.
9-8	INTTYPE1	0 1h 2h 3h	Interrupt Type of Interrupt Line 1. Indicates whether a buffer-full, RLOST, or BER interrupt, assigned to interrupt line 1, is currently pending. No interrupt. Interrupt caused by full buffer on CP/DCP specified by CPOFF1. RLOST interrupt generated by CP/DCP specified by CPOFF1. BER interrupt generated by CP/DCP specified by bits CPOFF1.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	CPOFF1	0 1h 2h 3h 4h 5h 6h 7h 8h 9h Ah Bh Ch Dh Eh Fh	CP Offset. Indicates for which DCP / CP the interrupt is pending, which is classified by INTTYPE1 and is assigned to interrupt line 1. DCP 0, CP A DCP 0, CP B DCP 1, CP A DCP 1, CP B DCP 2, CP A DCP 2, CP B DCP 3, CP A DCP 3, CP B DCP 4, CP A DCP 4, CP B DCP 5, CP A DCP 5, CP B DCP 6, CP A DCP 6, CP B DCP 7, CP A DCP 7, CP B

NOTE: Reading CPOFF1 will clear the bit generating the current interrupt from appropriate flag register (BERINTFL, RLOSTFL, or BFINTFL), except when in debug mode where reading CPOFF1 will have no effect on the flag registers.

In order to read INTTYPE1 and CPOFF1 simultaneously, always read this register using word or half-word but not using byte accesses.

24.4.14 Buffer Initialization Mode Register (HTU BIM)

This register enables special applications, where one CP is temporarily disabled, but after having re-enabled the CP, filling the buffer should not start back at its beginning, but should continue after the last element of the previous run.

Table 24-27 shows more details on the BIM usage.

Figure 24-27. Buffer Initialization Mode Register (HTU BIM) [offset = 3Ch]

31			16
Reserved			
R-0			
15	8	7	0
Reserved		BIM	
R-0		R/WP-0	

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset

Table 24-26. Buffer Initialization Mode Register (HTU BIM) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	BIM		Buffer Initialization Mode The BIM bits and the TMBx bits determine when a buffer is initialized, that means when its initial full address IFADDRx and its initial frame counter IFTCOUNT is used. When initializing (restarting) a buffer the information in the corresponding initial DCP RAM is loaded to a internal state machine but not to the current DCP RAM (CFADDRx, CFTCTx). The current DCP RAM is updated the first time when the first frame has finished. A buffer is initialized: • In circular buffer transfer mode (defined by TMBx) when the end of the buffer is reached. • When CPs are switched or enabled according to Buffer Initialization. The CPENA bits (2*x+1) and (2*x) are changed by write access to CPENA. For the first two rows of the table, the change of the CPENA bits could also be the result of the auto switch feature (as defined by TMBx). BIM bit x only affects DCP x (with x = 0,1,...or 7).

Table 24-27. Buffer Initialization

Case	Change of CPENA bits (2*x+1) and (2*x)		Action on buffer A or B (of DCP x)		
	Old state ⁽¹⁾	New state ⁽²⁾		BIM bit x = 0 (normal mode)	BIM bit x = 1 (special mode)
A	01	10	Switch from CP A to B	Next frame starts at the initial address of buffer B ⁽³⁾	Same as for BIM bit x = 0
B	10	01	Switch from CP B to A	Next frame starts at the initial address of buffer A ⁽³⁾	Same as for BIM bit x = 0
C	01	01	Stay at CP A	Write to CPENA bits (2*x+1) and (2*x) is ignored	Same as for BIM bit x = 0
E	10	10	Stay at CP B	Write to CPENA bits (2*x+1) and (2*x) is ignored	Same as for BIM bit x = 0
E	00	01	Enable CP A	Next frame starts at the initial address of buffer A	Next frame continues at the current address of buffer A
F	00	10	Enable CP B	Next frame starts at the initial address of buffer B	Next frame continues at the current address of buffer B
G	xx	11	Disable both CPs	Stop DCP x	Same as for BIM bit x = 0

(1) See read table of CPENA register ([Table 24-14](#)).

(2) See write table of CPENA register (Table 24-13).

⁽³⁾ This is regardless of whether the switch is done by a write access to CPENA or by the auto-switch feature.

NOTE: For cases E and F above, after the last frame of a buffer, the HTU sets CFTCTx to 0 and CFADDRx to the next address after the buffer. If the DCP was disabled during this state, then both CFTCTx and CFADDRx would contain invalid initialization values. Therefore, if a DCP should continue at its current address, then the software should use one of the following two procedures before it (re-) enables the DCP (as per [Table 24-27](#)):

1. If CFTCTx $\neq$ 0 then set BIM=1
 If CFTCTx = 0 then set
 BIM=0
2. If CFTCTx $\neq$ 0 then set
 BIM=1
 If CFTCTx = 0 then {set
 BIM=1;

```
set CFTCTx = IFTCOUNT;
set CFADDRx = IFADDRx}
```

But note that these procedures are only required for the cases E and F and not for all the other cases shown in [Table 24-27](#). Also, when a buffer reaches its end in circular mode, it uses the initial DCP information to restart independently of the BIM setting (assuming it is not temporarily disabled during CFTCTx = 0).

NOTE: Similarly, care needs to be taken when BIM is set to 1 and a DCP is enabled for the very first time. Also, in this case, CFTCTx and CFADDRx usually contain invalid initialization values. The software can either solve this by setting BIM = 0 for the first time or setting CFADDRx to IFADDRx and CFTCTx to IFTCOUNT before the DCP is enabled.

NOTE: If

- the HTUEN bit is changed to 1 after the HTU was disabled HTUEN = 0
- the CPENA bit pair is 01 or 10 (during this HTUEN change)

then the corresponding BIM bit will decide if the corresponding buffer continues at its initial or current address. Cases E and F in [Table 24-27](#) also apply for this situation. The software should use the procedures explained in the first note before setting HTUEN.

24.4.15 Request Lost Flag Register (HTU RLOSTFL)

Figure 24-28. Request Lost Flag Register (HTU RLOSTFL) [offset = 40h]

31	Reserved	16
	R-0	
15	CPRLFL	0
	R/W1CP-0	

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

Table 24-28. Request Lost Flag Register (HTU RLOSTFL) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	CPRLFL	0	CP Request Lost Flags
		0	No request was lost. Writing a 0 has no effect.
		1	If bit (2*x) is set, a request was lost on CP A of DCP x. If bit (2*x+1) is set, a request was lost on CP B of DCP x. Reading from INTOFFx in case of a RLOST interrupt clears the corresponding flag. The state of the flag bit can be polled even if RLINTENA is cleared.
			- Reading CPRLFL will not clear the flags or - Reading from INTOFFx clears the corresponding flag. - Writing a 1 clears the corresponding flag.

24.4.16 Buffer Full Interrupt Flag Register (HTU BFINTFL)

Figure 24-29. Buffer Full Interrupt Flag Register (HTU BFINTFL) [offset = 44h]

31	Reserved	16
	R-0	
15	BFINTFL	0
	R/W1CP-0	

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

Table 24-29. Buffer Full Interrupt Flag Register (HTU BFINTFL) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	BFINTFL	0	Buffer Full Interrupt Flags
		0	No buffer full condition is detected. Writing a 0 has no effect.
		1	If bit (2*x) is set, a buffer full condition on CP A of DCP x has been detected. If bit (2*x+1) is set, a buffer full condition on CP B of DCP x has been detected. The BFINTFL flag is set after the last frame finishes on the corresponding buffer regardless of whether the buffer is configured to one shot, circular or auto-switch mode. If BFINTFL is set in circular mode, then a circular overrun has occurred on the corresponding buffer. This can be used to indicate whether the buffer section after the frozen full address contains valid data or not. Reading from INTOFFx in case of a buffer-full interrupt clears the corresponding flag. The state of the flag bit can be polled even if the corresponding interrupt enable bit is cleared.
			- Reading BFINTFL will not clear the flags or - Reading INTOFFx will clear the corresponding flags or - Writing a 1 clears the corresponding flag.

24.4.17 BER Interrupt Flag Register (HTU BERINTFL)

A bus error interrupt results due to an address error or a timeout condition on the main memory access. A bus error will stop the frame transfer. Please see [Section 24.2.3](#).

Figure 24-30. BER Interrupt Flag Register (HTU BERINTFL) [offset = 48h]

31	Reserved	16
	R-0	
15	BERINTFL	0
	R/W1CP-0	

LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

Table 24-30. BER Interrupt Flag Register (HTU BERINTFL) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	BERINTFL	0	Bus Error Interrupt Flags
		0	No bus error condition is detected. Writing a 0 has no effect.
		1	If bit (2*x) is set, then a BER interrupt is pending on CP A of DCP x. If bit (2*x+1) is set, then a BER interrupt is pending on CP B of DCP x. The state of the flag bit can be polled even if BERINTENA is cleared.
			- Reading BERINTFL will not clear the flags or - Reading from INTOFFx in case of a BER interrupt clears the corresponding flag or - Writing a 1 clears the corresponding flag.

24.4.18 Memory Protection 1 Start Address Register (HTU MP1S)

This register configures the start address of memory protection region 1.

Figure 24-31. Memory Protection 1 Start Address Register (HTU MP1S) [offset = 4Ch]

31	STARTADDRESS1															16
R/WP-0																
15	STARTADDRESS1												2	1	0	
												0		0		
R/WP-0																

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset

Table 24-31. Memory Protection 1 Start Address Register (HTU MP1S) Field Descriptions

Bit	Field	Description
31-0	STARTADDRESS1	The start address defines at which main memory address the region begins. A memory protection error will be triggered, if the HTU accesses an address smaller than STARTADDRESS1 and the MPCS bit REG01ENA register is configured accordingly. The address is 32-bit aligned, so the 2 LSBs are not significant and will always read 0.

24.4.19 Memory Protection 1 End Address Register (HTU MP1E)

Figure 24-32. Memory Protection 1 End Address Register (HTU MP1E) [offset = 50h]

31	ENDADDRESS1															16
R/WP-0																
15	ENDADDRESS1													2	1	0
														0		0
R/WP-0																

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset

Table 24-32. Memory Protection 1 End Address Register (HTU MP1E) Field Descriptions

Bit	Field	Description
31-0	ENDADDRESS1	The end address defines at which address the region ends. A memory protection error will be triggered, if the HTU accesses an address bigger than ENDADDRESS1 and the register bit REG01ENA is configured accordingly. The address is 32-bit aligned, so the 2 LSBs are not significant and will always read 0. The effective end address is rounded up to the nearest word end address, that is, 0x200 = 0x203.

24.4.20 Debug Control Register (HTU DCTRL)

This register allows to create watch points on access to a certain location. It is intended to help debug the application execution during program development.

Figure 24-33. Debug Control Register (HTU DCTRL) [offset = 54h]

31	28	27	24	23	17	16	
Reserved		CPNUM		Reserved		HTUDBGS	
R-0		R-0		R-0		R/W1CS-0	
15						1	0
Reserved						DBREN	
R-0						R/WS-0	

LEGEND: R/W = Read/Write; R = Read only; W1CS = Write 1 in suspend mode to clear the bit; WS = Write in suspend mode only; -n = value after reset

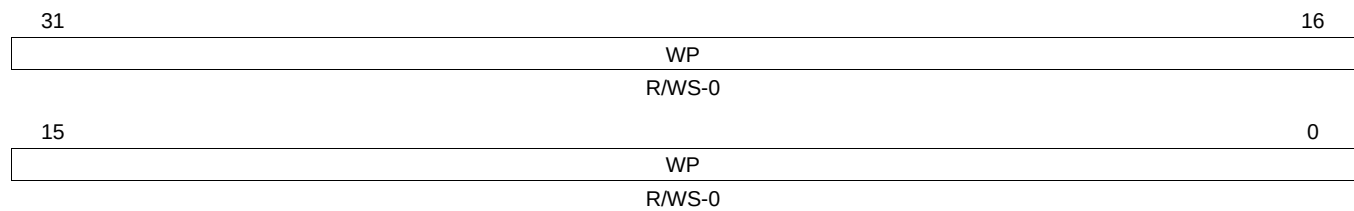
Table 24-33. Debug Control Register (HTU DCTRL) Field Descriptions

Bit	Field	Value	Description
31-28	Reserved	0	Reads return 0. Writes have no effect.
27-24	CPNUM	0	CP Number. These bit fields indicate the CP that should cause the watch point to match. CP A of DCP0
		1h	CP B of DCP0
		2h	CP A of DCP1
		3h	CP B of DCP1
		4h	CP A of DCP2
		5h	CP B of DCP2
		6h	CP A of DCP3
		7h	CP B of DCP3
		8h	CP A of DCP4
		9h	CP B of DCP4
		Ah	CP A of DCP5
		Bh	CP B of DCP5
		Ch	CP A of DCP6
		Dh	CP B of DCP6
		Eh	CP A of DCP7
		Fh	CP B of DCP7
23-17	Reserved	0	Reads return 0. Writes have no effect.
16	HTUDBGS	0	HTU Debug Status. When the main memory address is equal to the unique address defined by WPR, or lies in the specified range resulting from WMR, then the HTUDBGS is set. If in addition DBREN is set, then the application code execution will be stopped. A 1 must be written to this bit in order to clear it and to release the CPU from debug halting state. Read: No watch point condition was detected. Write: No effect.
		1	Read: A watch point condition was detected. Write: Clears the bit.
15-1	Reserved	0	Reads return 0. Writes have no effect.
0	DBREN		Debug Request Enable If a watch point matches and DBREN is set, then the application code execution will be stopped. This bit can only be set or cleared when in debug mode. This bit and all other bits of the DCTRL, WPR and WMR registers are reset by the test reset (nTRST) but not by the normal device reset.

24.4.21 Watch Point Register (HTU WPR)

This register defines the main memory address of the watch point.

Figure 24-34. Watch Point Register (HTU WPR) [offset = 58h]



LEGEND: R/W = Read/Write; WS = Write in suspend mode only; -n = value after reset

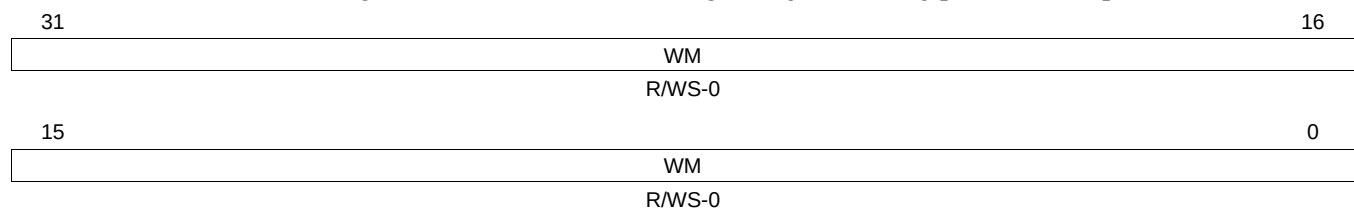
Table 24-34. Watch Point Register (HTU WPR) Field Descriptions

Bit	Field	Description
31-0	WP	Watch Point Register A 32-bit address can be programmed into this register as a watch point. The WPR register is used along with the Watch Mask Register (WMR). When the main memory address is equal to the unique address defined by WPR, or lies in the specified range resulting from WMR, then the HTUDBGS is set. If in addition DBREN is set, then the application code execution is stopped. This register can only be programmed during debug mode. This register and all other bits of the DCTRL and WMR registers are reset by the test reset (nTRST) but not by the normal device reset.

24.4.22 Watch Mask Register (HTU WMR)

This register defines a mask of the main memory address of the watch point. It can be used to define a memory range in conjunction with the WPR register.

Figure 24-35. Watch Mask Register (HTU WMR) [offset = 5Ch]



LEGEND: R/W = Read/Write; WS = Write in suspend mode only; -n = value after reset

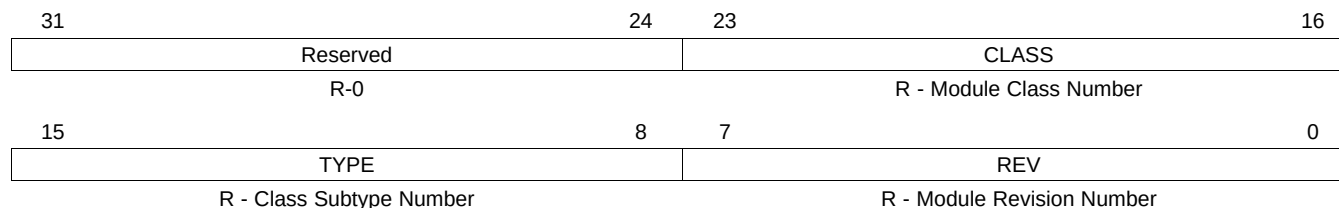
Table 24-35. Watch Mask Register (HTU WMR) Field Descriptions

Bit	Field	Description
31-0	WM	Watch Mask Register Setting a bit in the WMR register to 1 has the effect of masking the corresponding bit in of the main memory address, so that this bit is ignored for the address comparison. This register can only be programmed during debug mode. This register and all other bits of the DCTRL and WPR registers are reset by the test reset (nTRST) but not by the normal device reset.

24.4.23 Module Identification Register (HTU ID)

This register is for TI internal purposes and allows to keep track of the HTU module version on different devices.

Figure 24-36. Module Identification Register (HTU ID) [offset = 60h]



LEGEND: R = Read only; -n = value after reset

Table 24-36. Module Identification Register (HTU ID) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	CLASS		Module Class This field defines the module class number as read-only constant value for the HTU module. Writes have no effect.
15-8	TYPE		Subtype within a Class This field defines the subtype within a class as read-only constant value for the HTU module. Writes have no effect.
7-0	REV		Module Revision Number This field defines the module revision number as read-only constant value for the HTU module. Writes have no effect.

24.4.24 Parity Control Register (HTU PCR)

Figure 24-37. Parity Control Register (HTU PCR) [offset = 64h]

31													17	16
Reserved													COPE	
R-0													R/WP-0	
15					9	8	7					4	3	0
Reserved					TEST			Reserved				PARITY_ENA		
R-0					R/WP-0			R-0				R/WP-5h		

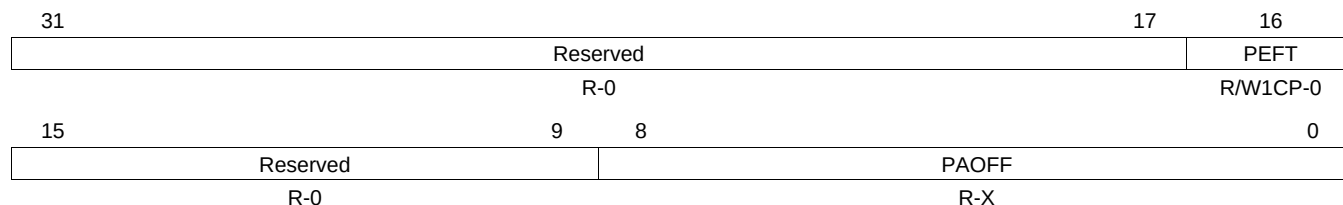
LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 24-37. Parity Control Register (HTU PCR) Field Descriptions

Bit	Field	Value	Description
31-17	Reserved	0	Reads return 0. Writes have no effect.
16	COPE	0	Continue on Parity Error The HTU performs parity checks every time it reads the RAM section of DCP x (with x = 0, 1,... or 7), before the next frame (of DCP x) is started. If a parity error is detected during this read access and if the parity check is enabled, then the frame will not be started and DCP x will be automatically disabled in the CPENA register. If a master different than the HTU (for example, CPU) reads the RAM section of DCP x and a parity error is detected during this read access, while the parity check is enabled, then the DCP x will automatically be disabled in the CPENA register. If a frame is active on DCP x during this read access, then in addition the element counter of DCP x is cleared and all new element transfers on DCP x are stopped and the active busy bit of DCP x is cleared.
		1	The difference to COPE = 0 is, that the data transfer on a active DCP continues after a parity error was detected on this DCP. So, neither the DCP with the parity error will be disabled nor the frame will be stopped.
15-9	Reserved	0	Reads return 0. Writes have no effect.
8	TEST	0	Test. When this bit is set, the parity bits are mapped into the peripheral RAM frame to make them accessible by the CPU. Parity bits are not memory-mapped.
		1	Parity bits are memory-mapped.
7-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	PARITY_ENA	5h	Enable/Disable Parity Checking. This bit field enables or disables the parity check on read operations and the parity calculation on write operations. If parity checking is enabled and a parity error is detected, then the PEFT flag is set, PAOFF is captured if it is not currently frozen and an interrupt is generated if it is enabled. Parity check is disabled.
		All Others	Parity check is enabled. Note: It is recommended to write Ah to enable error detection, to guard against single bit changes from flipping PARITY_ENA to a disable state.

24.4.25 Parity Address Register (HTU PAR)

Figure 24-38. Parity Address Register (HTU PAR) [offset = 68h]



LEGEND: R/W = Read/Write; R = Read only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset; X = undefined

Table 24-38. Parity Address Register (HTU PAR) Field Descriptions

Bit	Field	Value	Description
31-10	Reserved	0	Reads return 0. Writes have no effect.
16	PEFT	0 1	Parity Error Fault Flag. This bit is set, when the HTU detects a parity error and parity checking is enabled. No fault is detected. Fault is detected. Note: Once PEFT is set, a read access to the lower 16 bits or to the complete 32-bit HTUPAR register will clear the PEFT flag in non-debug mode. Another possibility to clear PEFT is to write a 1 to the PEFT bit.
15-9	Reserved	0	Reads return 0. Writes have no effect.
8-0	PAOFF		Parity Error Address Offset. This bit field holds the address of the first parity error, which is detected in the DCP RAM. PAOFF provides the offset address of the erroneous byte counted from the beginning of the DCP memory. This error address is frozen from being updated until a read access to the lower 16 bits or to the complete 32-bit HTUPAR register happens. During debug mode, this address is frozen even when read. Note: The Parity Error Address bits will not be reset, neither by PORRST nor by any other reset source.

24.4.26 Memory Protection Control and Status Register (HTU MPCS)

Figure 24-39. Memory Protection Control and Status Register (HTU MPCS) [offset = 70h]

31		28		27		24		
Reserved				CPNUM0				
R-0				R-0				
23		18		17		16		
Reserved				MPEFT1		MPEFT0		
R-0				R/W1CP-0		R/W1CP-0		
15		12		11		8		
Reserved				CPNUM1				
R-0				R-0				
7		6	5	4	3	2	1	0
Reserved		INT ENA01	ACCR01	REG01ENA	INT ENA0	ACCR0	REG0ENA	
R-0		R/WP-0	R/WP-0	R/WP-0	R/WP-0	R/WP-0	R/WP-0	R/WP-0

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; W1CP = Write 1 in privilege mode to clear the bit; -n = value after reset

Table 24-39. Memory Protection Control and Status Register (HTU MPCS) Field Descriptions

Bit	Field	Value	Description
31-28	Reserved	0	Reads return 0. Writes have no effect.
27-24	CPNUM0	0 1h 2h 3h 4h 5h 6h 7h 8h 9h Ah Bh Ch Dh Eh Fh	Control Packet Number for single memory protection region configuration. CPNUM0 holds the number of the CP, which has caused the first memory protection error when only one memory protection region is used. This number is not updated for multiple access violations until it is read by the CPU. During debug mode, CPNUM0 is frozen even when read. CP A of DCP0 CP B of DCP0 CP A of DCP1 CP B of DCP1 CP A of DCP2 CP B of DCP2 CP A of DCP3 CP B of DCP3 CP A of DCP4 CP B of DCP4 CP A of DCP5 CP B of DCP5 CP A of DCP6 CP B of DCP6 CP A of DCP7 CP B of DCP7
23-18	Reserved	0	Reads return 0. Writes have no effect.
17	MPEFT1	0 1	Memory Protection Error Fault Flag 1. This bit is set, when the HTU performs an access outside the region defined by the MP0S and MP0E and the MP1S and MP1E registers, when the access violates the rights defined by ACCR01, and when the REG01ENA bit is set. No fault detected. Writing a 0 has no effect. Fault detected. Writing a 1 will clear the bit.
16	MPEFT0	0 1	Memory Protection Error Fault Flag 0. This bit is set, when the HTU performs an access outside the region defined by the MP0S and MP0E registers, when the access violates the rights defined by ACCR, and when the REG0ENA bit is set. No fault detected. Writing a 0 has no effect. Fault detected. Writing a 1 will clear the bit.

Table 24-39. Memory Protection Control and Status Register (HTU MPCS) Field Descriptions (continued)

Bit	Field	Value	Description
15-12	Reserved	0	Reads return 0. Writes have no effect.
11-8	CPNUM1	Control Packet Number for single memory protection region configuration. CPNUM1 holds the number of the CP, which has caused the first memory protection error when only one memory protection region is used. This number is not updated for multiple access violations until it is read by the CPU. During debug mode, CPNUM1 is frozen even when read. 0 CP A of DCP0 1h CP B of DCP0 2h CP A of DCP1 3h CP B of DCP1 4h CP A of DCP2 5h CP B of DCP2 6h CP A of DCP3 7h CP B of DCP3 8h CP A of DCP4 9h CP B of DCP4 Ah CP A of DCP5 Bh CP B of DCP5 Ch CP A of DCP6 Dh CP B of DCP6 Eh CP A of DCP7 Fh CP B of DCP7	
7-6	Reserved	0	Reads return 0. Writes have no effect.
5	INTENA01	Interrupt Enable 01. This bit needs to be set when working with two memory-mapped regions and a error should be generated to the ESM module on an access violation. 0 Error signaling is disabled. 1 Error signaling is enabled.	
4	ACCR01	Access Rights 01. This bit defines the access rights for the HTU for accesses outside the region defined by the MP0S and MP0E and the MP1S and MP1E registers. 0 HTU read access is allowed but write access will be signaled. 1 Any access performed by the HTU is forbidden and will be signaled.	
3	REG01ENA	Region Enable 01. This bit needs to be set when working with two memory-mapped regions. REG0ENA must be cleared to 0 if this bit is set to a 1. Memory region 0 must be less than memory region 1. 0 The protection outside the memory region defined by the MP0S and MP0E and the MP1S and MP1E registers is not enabled. This means the HTU can access any implemented memory space. REG0ENA could still enabled to give protection outside the MP0S:MP0E region. 1 The protection outside the memory region defined by the MP0S and MP0E and the MP1S and MP1E registers is enabled. This means the HTU can perform any access within the regions, but if it attempts to perform a forbidden access outside of both of the regions (according to the ACCR01 configuration), the access is signaled by the MPEFT1 flag. The number of the CP, which has caused the memory protection error, is captured to CPNUM1 if it is not currently frozen and an error is generated if it is enabled.	
2	INTENA0	Interrupt Enable 0. This bit needs to be set when working with one memory-mapped region and a error should be generated to the ESM module on an access violation. 0 Error signaling is disabled. 1 Error signaling is enabled.	
1	ACCR	Access Rights 0. This bit defines the access rights for the HTU for accesses outside the region defined by the MP0S and MP0E registers for a single memory protection region configuration. 0 HTU read access is allowed but write access will be signaled. 1 Any access performed by the HTU is forbidden and will be signaled.	

Table 24-39. Memory Protection Control and Status Register (HTU MPCS) Field Descriptions (continued)

Bit	Field	Value	Description
0	REG0ENA	0	Region Enable 0
		0	The protection outside the memory region defined by the MP0S and MP0E registers is not enabled. This means the HTU can access any implemented memory space.
		1	The protection outside the memory region defined by the MP0S and MP0E registers is enabled. This means the HTU can perform any access within the region, but if it attempts to perform a forbidden access outside the region (according to the ACCR configuration), the access is signaled by the MPEFT0 flag, the number of the CP, which has caused the memory protection error, is captured to CPNUM0 if it is not currently frozen and an error is generated if it is enabled.

24.4.27 Memory Protection Start Address Register 0 (HTU MP0S)

This register configures the start address of memory protection region 0

Figure 24-40. Memory Protection Start Address Register 0 (HTU MP0S) [offset = 74h]

31																	16
STARTADDRESS0																	
R/WP-0																	
15											2	1	0				
STARTADDRESS0												0	0				
R/WP-0																	

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset

Table 24-40. Memory Protection 0 Start Address Register (HTU MP0S) Field Descriptions

Bit	Field	Description
31-0	STARTADDRESS0	The start address defines at which main memory address the region begins. A memory protection error will be triggered, if the HTU accesses an address smaller than STARTADDRESS0 and the MP0S register is configured accordingly. The address is 32-bit aligned, so the 2 LSBs are not significant and will always read 0.

24.4.28 Memory Protection End Address Register (HTU MP0E)

Figure 24-41. Memory Protection End Address Register (HTU MP0E) [offset = 78h]

31																	16
ENDADDRESS0																	
R/WP-0																	
15											2	1	0				
ENDADDRESS0												0	0				
R/WP-0																	

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset

Table 24-41. Memory Protection End Address Register (HTU MP0E) Field Descriptions

Bit	Field	Description
31-0	ENDADDRESS0	The end address defines at which address the region ends. A memory protection error will be triggered, if the HTU accesses an address bigger than ENDADDRESS0 and the register bit MP0E register is configured accordingly. The address is 32-bit aligned, so the 2 LSBs are not significant and will always read 0. The effective end address is rounded up to the nearest word end address, that is, 0x200 = 0x203.

24.5 Double Control Packet Configuration Memory

All bits marked "reserved" are implemented in RAM and will be initialized to unknown values after power on. Reserved locations can be written and read but should be written with 0 to ensure future compatibility. The HTU RAM can be cleared with the system RAM initialization function.

Table 24-42 provides a summary of the memory configuration. There are eight sets of DCP registers and eight sets of CF registers. The base address for the DCP registers is FF4E 0000h for HTU1 and FF4C 0000h for HTU2.

Table 24-42. Double Control Packet Memory Map

Offset	Acronym	Register Description	Section
00h	HTU DCP0 IFADDRA	Initial Full Address A Register	Section 24.5.1
04h	HTU DCP0 IFADDRB	Initial Full Address B Register	Section 24.5.2
08h	HTU DCP0 IHADDRCT	Initial N2HET Address and Control Register	Section 24.5.3
0Ch	HTU DCP0 ITCOUNT	Initial Transfer Count Register	Section 24.5.4
10h	HTU DCP1 IFADDRA	Initial Full Address A Register	Section 24.5.1
14h	HTU DCP1 IFADDRB	Initial Full Address B Register	Section 24.5.2
18h	HTU DCP1 IHADDRCT	Initial N2HET Address and Control Register	Section 24.5.3
1Ch	HTU DCP1 ITCOUNT	Initial Transfer Count Register	Section 24.5.4
:	:	:	
70h	HTU DCP7 IFADDRA	Initial Full Address A Register	Section 24.5.1
74h	HTU DCP7 IFADDRB	Initial Full Address B Register	Section 24.5.2
78h	HTU DCP7 IHADDRCT	Initial N2HET Address and Control Register	Section 24.5.3
7Ch	HTU DCP7 ITCOUNT	Initial Transfer Count Register	Section 24.5.4
100h	HTU CDCP0 CFADDRA	Current Full Address A Register	Section 24.5.5
104h	HTU CDCP0 CFADDRB	Current Full Address B Register	Section 24.5.6
108h	HTU CDCP0 CFCOUNT	Current Frame Count Register	Section 24.5.7
110h	HTU CDCP1 CFADDRA	Current Full Address A Register	Section 24.5.5
114h	HTU CDCP1 CFADDRB	Current Full Address B Register	Section 24.5.6
118h	HTU CDCP1 CFCOUNT	Current Frame Count Register	Section 24.5.7
:	:	:	
170h	HTU CDCP7 CFADDRA	Current Full Address A Register	Section 24.5.5
174h	HTU CDCP7 CFADDRB	Current Full Address B Register	Section 24.5.6
178h	HTU CDCP7 CFCOUNT	Current Frame Count Register	Section 24.5.7

24.5.1 Initial Full Address A Register (HTU IFADDRA)

Figure 24-42. Initial Full Address A Register (HTU IFADDRA)

31	16
IFADDRA	
R/WP-X	
15	0
IFADDRA	
R/WP-X	

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset; X = Unknown

Table 24-43. Initial Full Address A Register (HTU IFADDRA) Field Descriptions

Bit	Field	Description
31-0	IFADDRA	Initial Address of Buffer A in main memory. Initial (byte) address of buffer A placed in the main memory address range. Bits 0 and 1 are ignored by the logic, due to 32-bit alignment.

24.5.2 Initial Full Address B Register (HTU IFADDRB)

Figure 24-43. Initial Full Address B Register (HTU IFADDRB)

31	16
IFADDRB	
R/WP-X	
15	0
IFADDRB	
R/WP-X	

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset; X = Unknown

Table 24-44. Initial Full Address B Register (HTU IFADDRB) Field Descriptions

Bit	Field	Description
31-0	IFADDRB	Initial Address of Buffer B in main memory. Initial (byte) address of buffer B placed in the main memory address range. Bits 0 and 1 are ignored by the logic, due to 32-bit alignment.

24.5.3 Initial N2HET Address and Control Register (HTU IHADDRCT)

Figure 24-44. Initial N2HET Address and Control Register (HTU IHADDRCT)

31																24																																															
Reserved																																																															
R-0																																																															
23								22								21								20								19								18								17								16							
DIR								SIZE								ADDMH								ADDMF								TMBA								TMBB																							
R/WP-X								R/WP-X								R/WP-X								R/WP-X								R/WP-X								R/WP-X																							
15								13								12																								2								1								0							
Reserved								IHADDR																								Reserved																															
R-0								R/WP-X																								R-0																															

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

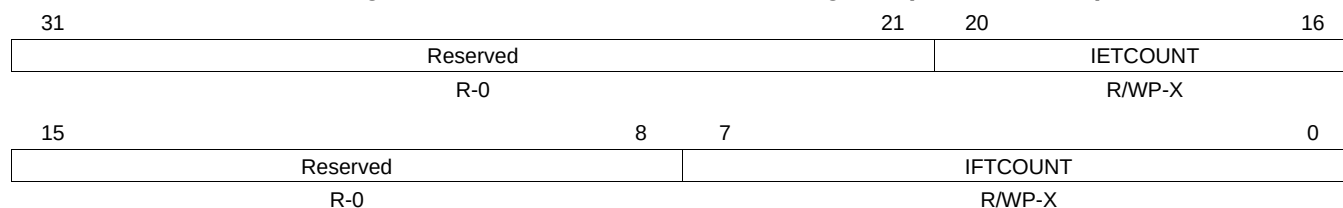
Table 24-45. Initial N2HET Address and Control Register (HTU IHADDRCT) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23	DIR	0 1	Direction of Transfer N2HET address is read and main memory address is written. Main memory address is read and N2HET address is written.
22	SIZE	0 1	Size of Transferred Data 32-bit transfer 64-bit transfer 64-bit transfer examples: If the N2HET address points to the N2HET instruction Control Field (CF), then the CF and Data Field (DF) will be transferred. If the N2HET address points to the Program Field (PF), then the PF and CF will be transferred.
21	ADDMH	0 1	Addressing Mode N2HET Address. This bit determines the N2HET address index from one to the next element of a frame. Increment by 16 bytes. Examples: If the initial N2HET address points to data field of instruction (n). Then the N2HET fields to be transferred by the elements of a frame are: data field of instruction (n), data field of instruction (n+1), data field of instruction (n+2) and so on. If the initial N2HET address points to control field of instruction (n), then the N2HET fields to be transferred by the elements of a frame are: control field of instruction (n), control field of instruction (n+1), control field of instruction (n+2) and so on. Increment by 8 bytes. This mode is intended to be used together with the 64-bit transfer size to load short N2HET instruction blocks into the N2HET RAM. So the sequence of transferred 64-bit elements could be: [PF and CF of instruction (n)], [DF and RF of instruction (n)], [PF and CF of instruction (n+1)], [DF and RF of instruction (n+1)] and so on.
20	ADDMF	0 1	Addressing Mode Main Memory Address Post-increment Note: When post-increment is selected the HTU will automatically increment by 4 bytes for a 32-bit data size and by 8 bytes for a 64-bit data size. Constant
19-18	TMBA	0 1h 2h-3h	Transfer Mode for Buffer A One-Shot buffer mode Circular buffer mode Auto Switch mode
17-16	TMBB	0 1h 2h-3h	Transfer Mode for Buffer B One-Shot buffer mode Circular buffer mode Auto Switch mode

Table 24-45. Initial N2HET Address and Control Register (HTU IHADDRCT) Field Descriptions (continued)

Bit	Field	Value	Description
15-13	Reserved	0	Reads return 0. Writes have no effect.
12-2	IHADDR		Initial N2HET Address The initial N2HET Address points to the N2HET field, which is the first element of the frame. The N2HET address (bits 12:2) increments by 1 for each 32-bit N2HET field and starts with 0 at the first 32-bit field in the N2HET RAM. Note: When the HTU addresses the N2HET RAM it uses only the number of address bits required for the actual N2HET RAM size. If the N2HET address exceeds the actual N2HET RAM size, the unused MSB bits of the address will be ignored and the address rolls over to the start of the N2HET RAM.
1-0	Reserved	0	Reads return 0. Writes have no effect.

24.5.4 Initial Transfer Count Register (HTU ITCOUNT)

Figure 24-45. Initial Transfer Count Register (HTU ITCOUNT)


LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

Table 24-46. Initial Transfer Count Register (HTU ITCOUNT) Field Descriptions

Bit	Field	Value	Description
31-21	Reserved	0	Reads return 0. Writes have no effect.
20-16	IETCOUNT		Initial Element Transfer Count Defines the number of element transfers.
15-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	IFTCOUNT		Initial Frame Transfer Count Defines the number of frame transfers.

24.5.5 Current Full Address A Register (HTU CFADDRA)

Figure 24-46. Current Full Address A Register (HTU CFADDRA)

31	CFADDRA	16
	R/WP-X	
15	CFADDRA	0
	R/WP-X	

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset; X = Unknown

Table 24-47. Current Full Address A Register (HTU CFADDRA) Field Descriptions

Bit	Field	Description
31-0	CFADDRA	Current (byte) Address of Buffer A The current main memory address register is updated at the end of each frame. Therefore it points to the start address of the frame, which is the next to transfer, if currently no frame is transferred on this DCP. For an ongoing frame transfer, it points to the start address of this frame. After the last element of a buffer was transferred it will point to the buffer end address plus 0x4. The main purpose of the current full address registers for buffer A and buffer B (see next section) is to enable the software to find out the recently transferred element in the frozen buffer while the address of the active buffer increments. Note: A frame can be automatically stopped if any of the events listed in Conditions for Frame Transfer Interruption happens. If a frame is stopped before it could complete, then the current full address register is not updated and it will point to the start of the bad frame after the DCP was automatically disabled.

To transfer the first frame of buffer x, the information in the corresponding initial DCP RAM (IFADDRx, IHADDRCT, ITCOUNT) is loaded to an internal state machine but not to the current DCP RAM (CFADDRx, CFTCTx).

This is valid for all of the following modes:

- Buffer x has reached its end in circular mode and rolls back to its start address.
- CP x is enabled by a CPENA access (and corresponding BIM bit is 0).
- A CPENA access or auto-switch mode causes a switch from CP y to CP x.

This means after starting the transfer to/from buffer x, CFADDRx and CFTCTx is not updated before the end of the first frame. So before the software switches from CP y to CP x using a write access to the CPENA register, it needs to initialize CFADDRx, CFTCTx. This allows the software to find out if the next request on CP x after the switching to CP x was delayed or never occurring.

24.5.6 Current Full Address B Register (HTU CFADDRB)

Figure 24-47. Current Full Address B Register (HTU CFADDRB)

31	CFADDRB	16
	R/WP-X	
15	CFADDRB	0
	R/WP-X	

LEGEND: R/W = Read/Write; WP = Write in privilege mode only; -n = value after reset; X = Unknown

Table 24-48. Current Full Address B Register (HTU CFADDRB) Field Descriptions

Bit	Field	Description
31-0	CFADDRB	Current (byte) Address of Buffer B The current main memory address register is updated at the end of each frame. Therefore it points to the start address of the frame, which is the next to transfer, if currently no frame is transferred on this DCP. If currently a frame is transferred, then it points to the start address of this frame. After the last element of a buffer was transferred it will point to the buffer end address plus 0x4. The main purpose of the current full address registers for buffer A and buffer B (see next section) is to enable the software to find out the recently transferred element in the frozen buffer while the address of the active buffer increments. Note: A frame can be automatically stopped if any of the events listed in Conditions for Frame Transfer Interruption happens. If a frame is stopped before it could complete, then the current full address register is not updated and it will point to the start of the bad frame after the DCP was automatically disabled.

To transfer the first frame of buffer x, the information in the corresponding initial DCP RAM (IFADDRx, IHADDRCT, ITCOUNT) is loaded to an internal state machine but not to the current DCP RAM (CFADDRx, CFTCTx).

This is valid for all of the following modes:

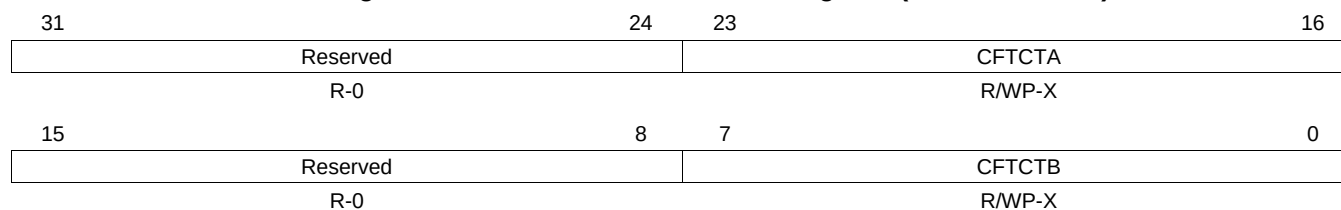
- Buffer x has reached its end in circular mode and rolls back to its start address.
- CP x is enabled by a CPENA access (and corresponding BIM bit is 0).
- A CPENA access or auto-switch mode causes a switch from CP y to CP x.

This means after starting the transfer to/from buffer x, CFADDRx and CFTCTx is not updated before the end of the first frame. So before the software switches from CP y to CP x using a write access to the CPENA register, it needs to initialize CFADDRx, CFTCTx. This allows the software to find out if the next request on CP x after the switching to CP x was delayed or never occurring.

24.5.7 Current Frame Count Register (HTU CFCOUNT)

The current frame count register enables the software to find out the recent frame in the buffer while the counter of the active buffer decrements.

Figure 24-48. Current Frame Count Register (HTU CFCOUNT)



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privilege mode only; -n = value after reset; X = Unknown

Table 24-49. Current Frame Count Register (HTU CFCOUNT) Field Descriptions

Bit	Field	Value	Description
31-24	Reserved	0	Reads return 0. Writes have no effect.
23-16	CFTCTA		Current Frame Transfer Count for CP A. It is updated at the end of each frame.
15-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	CFTCTB		Current Frame Transfer Count for CP B. It is updated at the end of each frame.

24.6 Examples

24.6.1 Application Examples for Setting the Transfer Modes of CP A and B of a DCP

Table 24-50. Application Examples for Setting the Transfer Modes of CP A and B of a DCP

CP A	CP B	
One shot	Not used	Buffer A can be used as a "one shot" buffer. A buffer full interrupt enabled for CP A can signal reaching the end of the buffer.
Auto switch	One shot	Can double the buffer size for a "one shot" buffer. A buffer full interrupt enabled for CP B can signal reaching the end of the buffer.
Circular	Circular	The CPU can switch the buffers at arbitrary times. It will fill or read the frozen buffer during the other buffer is filled or read by the HTU. Interrupts are not required for this case.
Auto switch	Auto switch	Buffer full interrupts (enabled for CP A and B) signal when the end of a buffer is reached. After one buffer is completed the according CPU interrupt routine will read or refill this buffer. At the same time the other buffer is read or filled by the HTU. Here the time when the buffer must be read is determined by the time of the interrupt (determined by the frequency of the N2HET transfer requests).

24.6.2 Software Example Sequence Assuming Circular Mode for Both CP A and B

The example assumes the N2HET address to be read and the main memory address to be written.

- I1 CPU initializes initial DCP: IFADDRA, IFADDRB, IHADDRCT, ITCOUNT
- I2 CPU clears current DCP: CFADDRA, CFADDRB, CFTCTA, CFTCTB
- I3 CPU clears BFINTFL flag of CP A and B
- I4 Enable CP A with the CPENA register. Now the HTU fills buffer A

After some time the CPU intends to read buffer A:

- A1 CPU enables CP B and disables CP A by writing to the CPENA register. After this switch, the HTU fills buffer B. Filling buffer B starts with its initial full address and initial frame counter.
- A2 CPU waits for CP A busy bit equals 0
- A3 Optional: CPU verifies that the CP A request lost flag is not set. The bus error flag of CP A could also be checked.
- A4 CPU reads the frozen CFTCTA, which indicates the fill level in the buffer
- A5 CPU sets current CP A (CFTCTA and/or CFADDRA) to 0. This allows to find out if any request has happened during the next time buffer A is active.
- A6 CPU reads BFINTFL flag of buffer A
- A7 CPU clears the BFINTFL flag of buffer A. This is an initialization for the next time buffer A is used.
- A8 CPU reads valid values of frozen buffer A. After reading the CPU does not need to clear the frozen buffer A.

After some time the CPU intends to read buffer B:

- | | |
|----|--|
| B1 | CPU enables CP A and disables CP B by writing to the CPENA register. After this switch, the HTU fills buffer A. Filling buffer A starts with its initial full address and initial frame counter. |
| B2 | CPU waits for CP B busy bit equals 0 |
| B3 | Optional: CPU verifies that the CP B request lost flag is not set. The bus error flag of CP B could also be checked. |
| B4 | CPU reads the frozen CFTCTB, which indicates the fill level in the buffer |
| B5 | CPU sets current CP B (CFTCTB and/or CFADDRB) to 0. This allows to find out if any request has happened during the next time buffer B is active. |
| B6 | CPU reads BFINTFL flag of buffer B |
| B7 | CPU clears the BFINTFL flag of buffer B. This is an initialization for the next time buffer B is used. |
| B8 | CPU reads valid values of frozen buffer B. After reading the CPU does not need to clear the frozen buffer B. |

After some time the CPU intends to read buffer A:

A1) ... see above...

NOTE: The buffer full interrupt doesn't need to be enabled. The BFINTFL flag is used to indicate a circular overrun of the buffer. If the BFINTFL flag is set, also the buffer section after the frozen full address could be read.

Steps A3 and B3 in the example sequence above imply that request lost interrupts are disabled. The example below assumes that request lost interrupts are enabled.

Request lost detection with interrupt enabled.

24.6.3 Example of an Interrupt Dispatch Flow for a Request Lost Interrupt

- A request lost occurs and the interrupt routine starts.
- Reading INTOFFx.INTYPEx shows that RLOSTFL is the interrupt source.
- Reading INTOFFx.CPOFFx = Ah shows that DCP 5 / CP A has caused the RLOSTFL interrupt. The hardware automatically clears bit (2·5+0) in RLOSTFL.
- Reading RLOSTFL= 84h shows that also another request lost event happened on DCP 1 / CP A [bit (2·1+0)] and on DCP 3 / CP B [bit (2·3+1)] at the same time or after the request lost occurred on DCP 5 / CP A.
- Writing back 84h to RLOSTFL clears bits 2 and 7 and the according pending interrupts.